



Mitsubishi Electric Industrial Robot

RV-5AS

Instruction Manual

Robot Arm Setup and Maintenance



Safety Precautions

Always read the following precautions and the separate "Collaborative Robot Safety Manual" before starting use of the robot to learn the required measures to be taken.

■ Basic precautions and important points when using collaborative robots

Collaborative robots have a variety of safety functions, so unlike conventional industrial robots, they can work in the same space as humans without being separated by a machine guard.

Even though collaborative robots are equipped with safety functions, it does not guarantee that they will not cause injury. To prevent injury, customers who use robots, machine manufacturers who construct robot systems, and system builders such as system integrators must select and use functions correctly. Peripherals must be designed, manufactured, set, programmed, maintained, and inspected appropriately. Before working with collaborative robots, conduct risk assessments, check that no risk is present, and ensure all required documentation is in order.

It is vitally important that all the points mentioned above are fully understood in order to work with collaborative robots safely and without incident. When using collaborative robots, always keep in mind that they may move unexpectedly.

Revision history

Date of Print	Instruction Manual No.	Revision Details
2020-04-10	BFP-A3729	• First print
2020-07-30	BFP-A3729-A	• Revised the guidelines for unpacking the robot.
2021-08-31	BFP-A3729-B	• Revised "3.1 Installing the solenoid valve set".
2022-02-24	BFP-A3729-C	• Software version C2d supported. Added descriptions to "2.3.2 Turning ON the control power". Added the parameter HNDCHK to "Table 2-4: Hand parameter".
2023-09-14	BFP-A3729-D	• Added "Appendix 1: How to take out the battery at the disposal of the robot".
2024-04-05	BFP-A3729-E	• Updated branch information. • Corrected other mistakes.

*Introduction

Thank you for purchasing the Mitsubishi industrial robot.

This instruction manual explains the method of unpacking, installation and maintenance and inspection of the robot arm.

Always read through this manual before starting use to ensure correct usage of the robot.

The information contained in this document has been written to be accurate as much as possible.

Please interpret that items not described in this document "cannot be performed."

This document explains for the following robot type.

<Robot type>

- RV-5AS

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1 Before starting use

This chapter explains how to use this manual. It also contains information on safety and the basic symbols used in it. Moreover, handling and operation of a teaching pendant (T/B) are described based on R32TB in instruction manuals. If using other T/B, such as R56TB, refer to a supplied instruction manual of the T/B.

1.1 Using the instruction manuals

1.1.1 Details of each instruction manual

The contents and purposes of the documents enclosed with this product are shown below. Use these documents according to the application.

Manual name	Description	Manual No.
Collaborative Robot Safety Manual	Describes the common precautions and safety measures to be taken for robot handling, system design and manufacture to ensure safety of the operators involved with the robot. Read this manual first.	BFP-A3733
Hello ASSISTA Quick Set-up Guide	Describes procedures including unpacking, installation, programming using RT VisualBox, and operation of the robot.	BFP-A3715
Standard Specifications Manual	Describes the standard specifications, option configuration, and maintenance parts of the product. This also describes safety and technical precautions when incorporating the robot.	BFP-A3727
Robot Arm Setup and Maintenance (This manual)	Describes the procedures (unpacking, transport, installation, and operation check), maintenance, and inspection to use the robot arm.	BFP-A3729
Controller Setup and Maintenance	Describes the procedures (unpacking, transport, and installation), maintenance, and inspection to use the robot controller.	BFP-A3731
Collaborative Robot: Detailed explanations of functions and operations	Describes functions specific to the collaborative robot.	BFP-A3735
Detailed explanations of functions and operations ^{Note1)}	Describes details on functions and operation including information of how to use functions, MELFA-BASIC VI commands used in programs, how to connect external input/output devices, and parameters. The information of functions and operation refers to the FR series. For unsupported functions, refer to the Collaborative Robot: Detailed explanations of functions and operations.	BFP-A3478
Troubleshooting ^{Note1)}	Describes a cause and solution corresponding to an error number.	BFP-A3480
Tracking Function ^{Note1)}	Describes the specifications, functions, and usage of the conveyor tracking.	BFP-A3520
GOT Direct Connection Extended Function ^{Note1)}	Describes the data configuration of the memory between the GOT and the robot, monitoring, and operating procedures.	BFP-A3546
Ethernet Function ^{Note1)}	Describes how to enable the communications with a personal computer on Ethernet by using the TCP/IP protocol.	BFP-A3379

Note1) Instances where the CR800-D controller is mentioned also refer to the CR800-05VD.

1.1.2 Symbols used in this manual

Symbols used in this manual are explained in [Table 1-1](#).

Learn the meaning of these symbols before reading this instruction manual.

Table 1-1: Symbols in instruction manual

Terminology	Symbol	Meaning
Symbol	 DANGER	Precaution indicating cases where there is a risk of operator fatality or serious injury if handling is mistaken. Always observe these precautions to safely use the robot.
	 WARNING	Precaution indicating cases where the operator could be subject to fatalities or serious injuries if handling is mistaken. Always observe these precautions to safely use the robot.
	 CAUTION	Precaution indicating cases where operator could be subject to injury or physical damage could occur if handling is mistaken. Always observe these precautions to safely use the robot.
	[JOG]	Square brackets ([]) are used to represent operation keys.
	[RESET] + [EXE] (A) (B)	This indicates to press the (B) key while holding down the (A) key. In this example, the [RESET] key is pressed while holding down the [EXE] key.
	T/B	This indicates the teaching pendant. Descriptions in this manual are based on R32TB.

2 Unpacking to Installation

2.1 Confirming the product

The standard configuration of the robot arm, part of the purchased product, is shown in [Table 2-1](#).

Confirm the parts.

Users who have purchased optional products should refer to the separate "Standard Specifications Manual".

Table 2-1: Standard configuration

No.	Part name	Type	Qty.	Remarks
1	Robot arm	-	1 unit	-
2	Installation bolts	M8 × 35	4 pcs.	For robot arm installation Included with the robot controller
3	Spring washer for installation bolts	For M8	4 pcs.	
4	Plain washer for installation bolts	For M8	4 pcs.	

2.2 Installation



CAUTION

Read the separate Collaborative Robot Safety Manual and following precautions thoroughly before using this product. Take precautions as necessary.

2.2.1 Unpacking

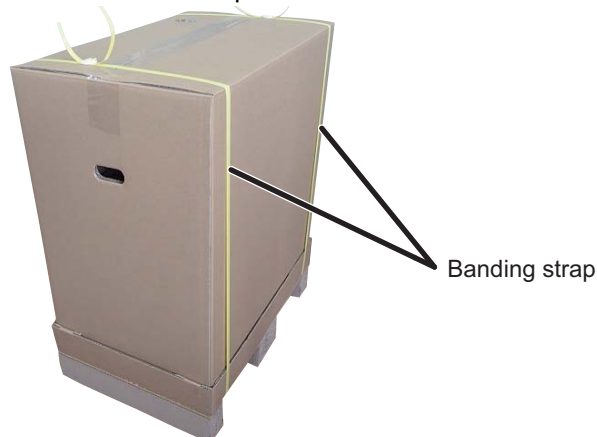
The robot is shipped from the factory in cardboard packing. For details on transportation of the robot arm, refer to ["2.2.2 Transportation procedure \(with two people\)"](#).



CAUTION

Always unpack the robot at a flat place. The robot could tilt over if unpacked at an unstable place.

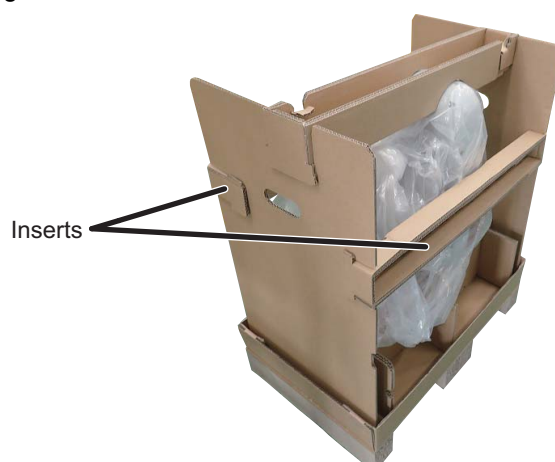
- 1) Cut the banding straps with scissors and open the box.



2) Pull the box up and remove it.



3) Remove the left and right inserts.



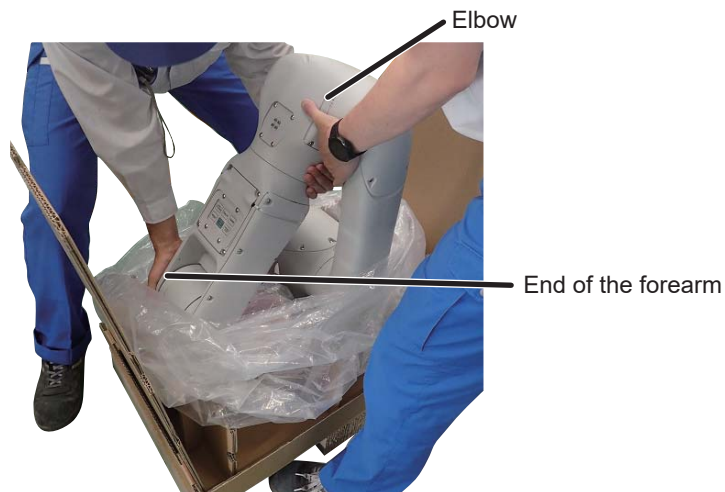
4) Remove the top inserts.



5) Open the plastic bag covering the robot.



6) Remove the robot from the box. When lifting the robot, one person should hold the end of the forearm and elbow, while another person holds the area under the elbow.



Mass: 32kg



CAUTION

Place the robot on a flat surface. Fix the robot in place or have somebody support it during installation. External forces may cause the robot to fall if the robot is not fixed in place or supported during installation.



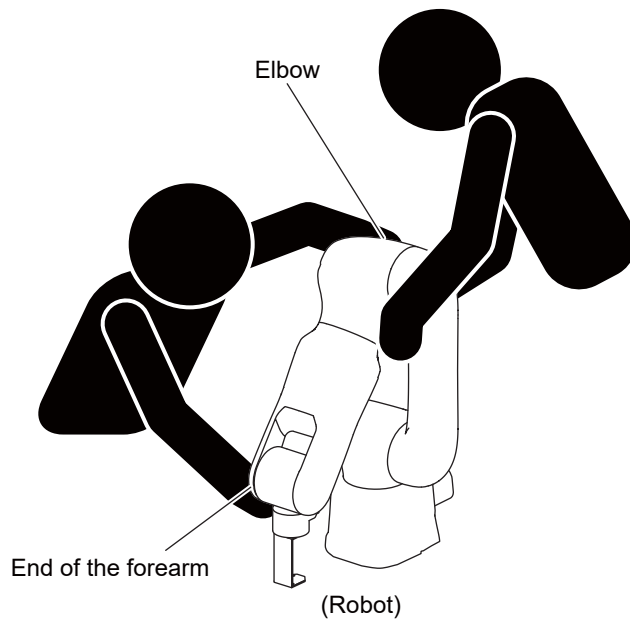
- 7) Remove the support bracket for the mechanical interface once the robot has been installed.



CAUTION

The support bracket and packaging are required when relocating the robot. Keep them in a safe place.

2.2.2 Transportation procedure (with two people)



Mass: 32kg

Fig.2-1: Transportation of robot arm (Transportation by people)

- 1) The robot must be transported by two workers. Use a cart or other device for transporting the robot near the installation place. The following instructions for carrying the robot are applicable only when the robot is transferred onto the stand or another cart, or moved for positioning.
- 2) Transport the robot in the same posture that it came in, with one person holding the end of the forearm and elbow, and another person holding the area under the elbow. Never hold the robot by its cover. Holding the robot by its cover will cause an accident as the robot may fall over, drop, or the cover may become damaged. Do not apply force to the cover or subject the robot to large impacts during transportation. Ensure that the machine cable does not get caught on anything or get pulled during transportation.
- 3) Transfer the robot slowly. Be careful not to get injured with the edge of the robot arm.
- 4) Before moving the robot to a new location, re-attach the support bracket to the mechanical interface, then secure the robot in the packaging that it came in. Transporting the robot without attaching it to the support bracket may cause damage to peripherals. Furthermore, the robot may become unstable due to an unfavorable center of gravity and cause an accident.



CAUTION

Do not hold the cover of the robot for carrying it.



CAUTION

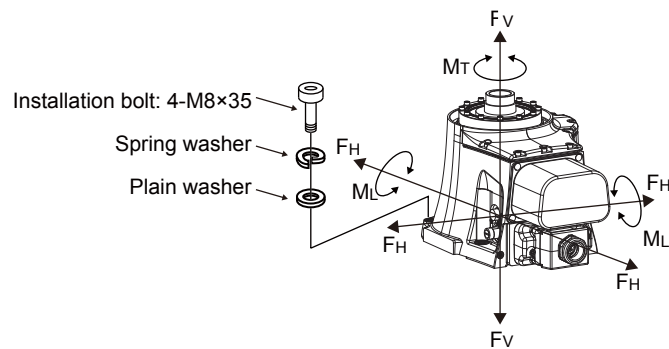
For secondary transportation, adjust the posture of each axis of the robot as specified in [Table 2-2](#).

Table 2-2: Transportation posture

Axis	Angle
J1	0°
J2	0°
J3	145°
J4	0°
J5	35°
J6	-90°

2.2.3 Installation procedures

The installation procedure of the robot arm is shown below.



<Bottom view>

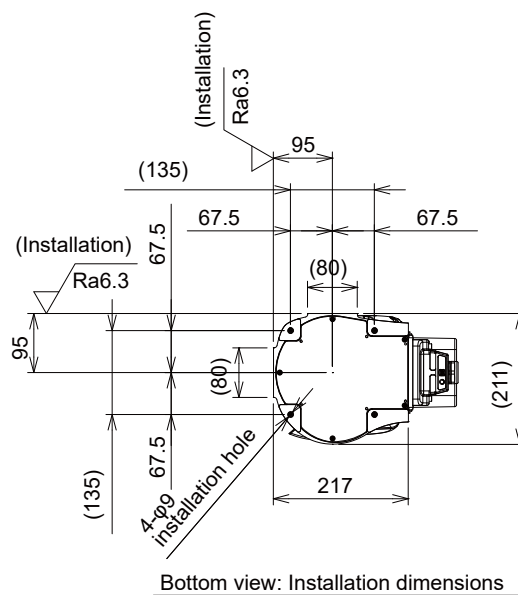


Fig.2-2: Installation dimensions

- 1) The base of the robot has already been machined flat. There are four installation holes ($\phi 9$) on the base of the robot. Secure the robot into place using these holes with the included parts below. (Recommended tightening torque: 26.5 N•m)
 - Installation screws (M8 hex socket screws)
 - Spring washers for installation screws
 - Plain washers for installation screws
- 2) Install the robot on a level surface.
- 3) It is recommended that the surface roughness of the table onto which the robot is to be installed by 6.3 Ra or more. If the installation surface is rough, the contact with the table will be poor, and positional deviation could occur when the robot moves.
- 4) When installing, use a common table to prevent the position of the devices and jigs subject to robot work from deviating.
- 5) The installation surface must have sufficient strength to withstand the arm reaction during operation, and resistance against deformation and vibration caused by the static (dynamic) load of the robot arm and peripheral devices, etc.
- 6) The installation surface must have sufficient strength to withstand the arm reaction during moving the robot at high speed.

Table 2-3: Strength of the installation side (reference)

Item	Unit	Value
Tilt moment: M_L	N•m	325
Torsional moment: M_T	N•m	275
Horizontal direction translation force: F_H	N	700
Vertical direction translation force: F_V	N	1,010

⚠ CAUTION

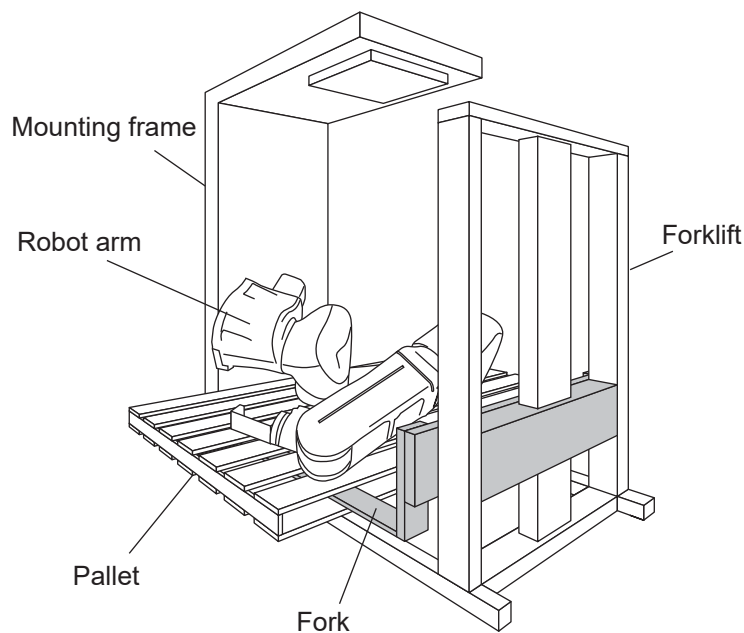
During installation, ensure that there is enough space at the rear of the robot to connect the machine cable. Refer to "Outside dimension • Operating range diagram" in the Standard Specifications Manual for the dimensions of the space required to connect the machine cable.

⚠ CAUTION

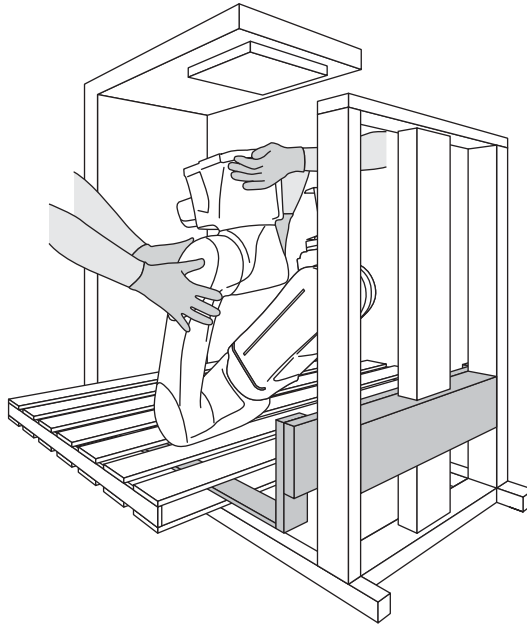
Do not install the robot arm in areas where direct sunlight is present or heat is generated from lighting.
A rise in the temperature of the robot arm surface may cause an error.

2.2.4 Ceiling mounting procedure

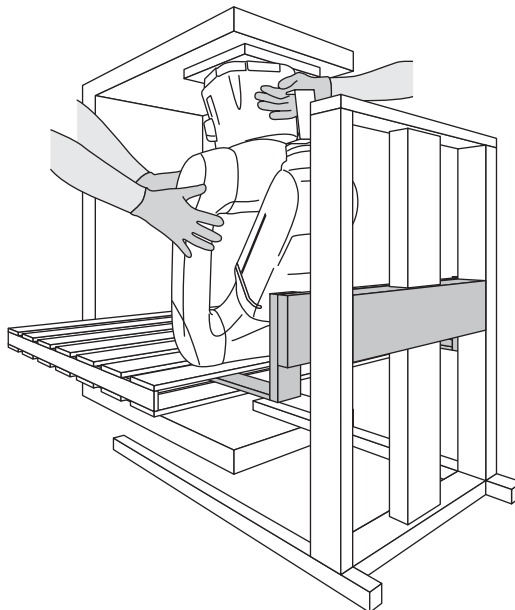
- 1) Use three people to assist with the installation of the robot when mounting it to a ceiling. Transport the robot to a position beneath the installation area. Refer to ["2.2.2 Transportation procedure \(with two people\)"](#) before moving the robot to this position.
- 2) Insert the forks of a forklift into a pallet, then place the robot on the pallet. To protect the acrylic cover, set the robot down on the side of its upper arm. Lay a mat or similar cushioning over the pallet to protect the robot from scratches.



- 3) Face the robot elbow down. Use two people to support the robot, then face the base upward towards the installation area.



- 4) While the two people support the robot, have a third person lift the pallet up with the forklift so that the base of the robot comes into contact with the installation area.



- 5) There are four installation holes ($\phi 9$) on the base of the robot. Secure the robot into place using these holes with the included parts below. (Recommended tightening torque: 26.5 N•m)
- Installation screws (M8 hex socket screws)
 - Spring washers for installation screws
 - Plain washers for installation screws

2.2.5 Grounding procedures

(1) Grounding methods

- 1) As indicated in Fig. 2-3, there are three ways of grounding (earthing) the RV-5AS. The most preferable of the three is "a". Do not use "c". For information on grounding (earthing) the robot controller, refer to the separate Instruction Manual "Controller Setup and Maintenance".
- 2) Use Class D grounding (grounding resistance 100Ω or less). Dedicated grounding separated from the other devices should be used.
- 3) Use a AWG#11(4.2mm²) or more stranded wire for the grounding wire. The grounding point should be as close to the robot arm and controller as possible, and the length of the grounding wire should be short.

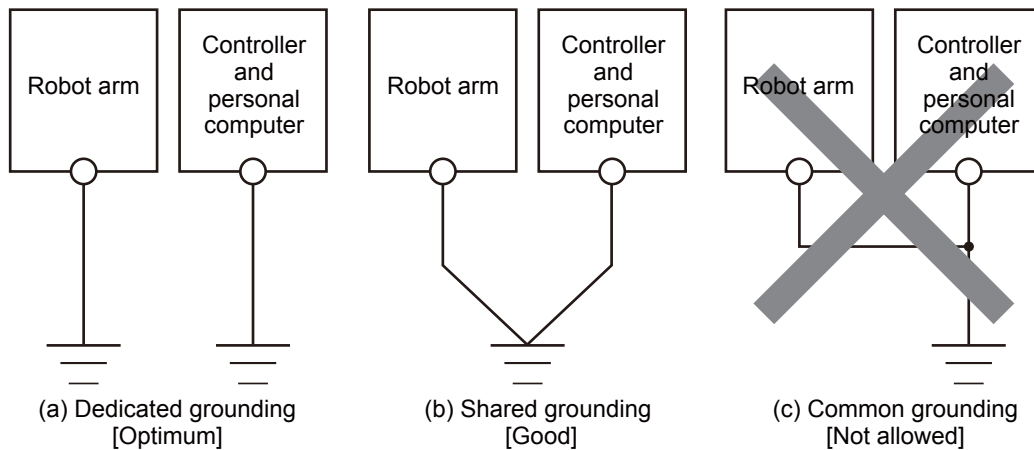


Fig.2-3: Grounding methods

(2) Grounding procedures

- 1) Prepare the grounding cable (AWG#11(4.2mm²) or more) and robot side installation screw and washer.
- 2) If there is rust or paint on the grounding screw section (A), remove it with a file, etc.
- 3) Connect the grounding cable to the grounding screw section.

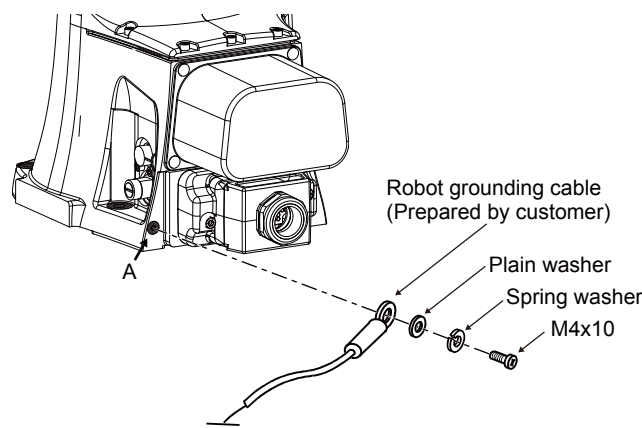


Fig.2-4: Connecting the grounding cable

2.2.6 Connecting with the controller

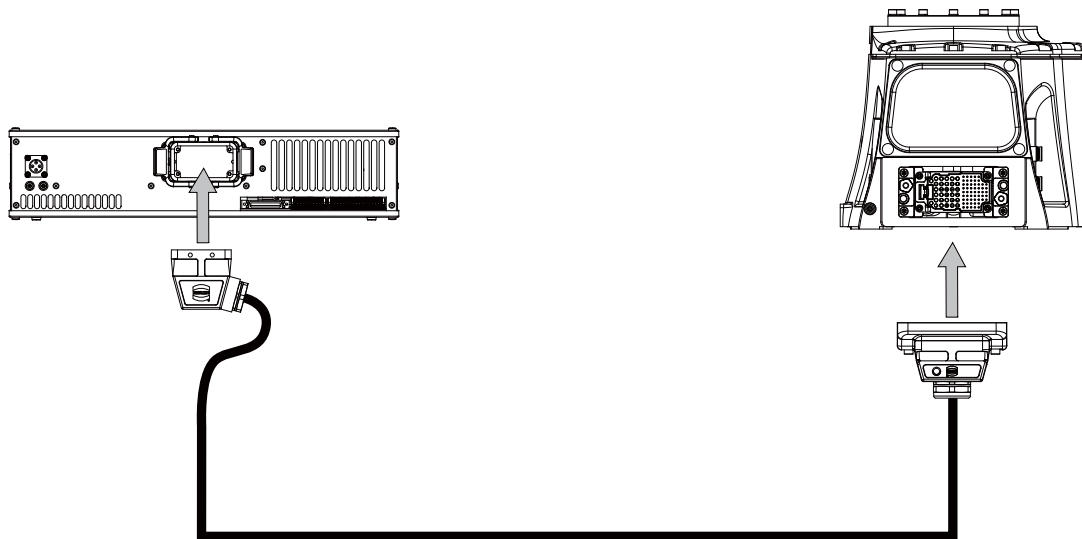


Fig.2-5: Connecting the machine cables

Carry out the following procedure after installing the controller referring to the separate "Controller Setup and Maintenance" manual.

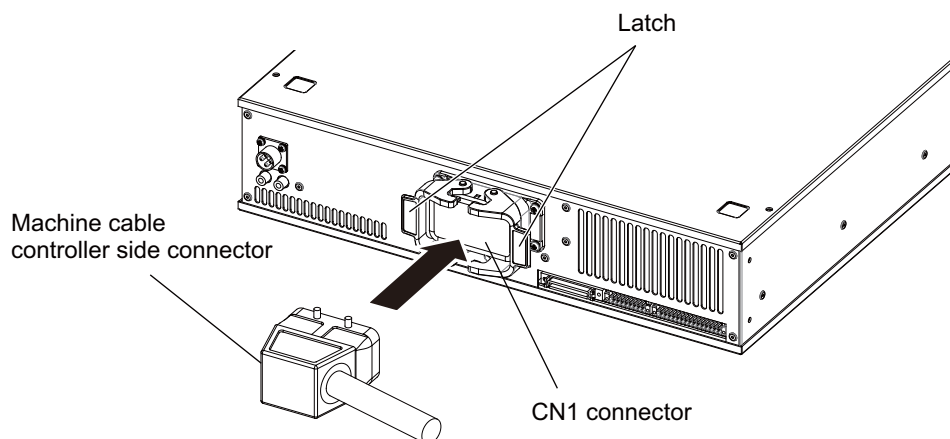


Fig.2-6: Connecting with the controller

- 1) Make sure that the power of the controller is turned OFF.
- 2) Connect the controller side connector of the machine cable to CN1 connector on the rear side of the controller.
- 3) To fix the inserted connector, close the latches of the CN1 connector.
- 4) The connection method is the same for the optional machine cables (replaceable type).



CAUTION

When connecting and disconnecting the connector, be careful not to get your hand pinched.



CAUTION

When installing or removing the connector, to the connector of the other party in parallel, install or remove.
If load strong against one side is applied, the connector pin may be damaged and it may not be connected securely.



CAUTION

To disconnect the cable, open the latches, hold the connector part, and pull out the cable. Pulling the cable part may cause the cable to come off or break.



CAUTION

Take special care to the leading of the connection cable. If the cable is pulled with force or bent excessively, wires could break or the connector could be damaged.



CAUTION

Connect the machine cable at the place without the effect of the dust or oil mist.

2.3 Confirming the operation

After the robot is connected to the controller, the origin data is automatically written.

Then, manually move the robot using the T/B to confirm that the operation is correct.

Moving the robot manually is called "jog operation". This operation includes the JOINT jog that moves each axis, the XYZ jog that moves along the base coordinate system, the TOOL jog that moves along the tool coordinate system, the Work jog that moves along the work coordinate system, and the CYLINDER jog that moves along the circular arc.

This operation is carried out while pressing the deadman switch on the back of the T/B.

Note) The figure of the robot which indicated to the explanation page in each jog mode is an example.



CAUTION

The robot will move during this operation. Make sure that there are no operators near the robot, and that there are no obstacles, such as tools, in the robot operation range.



CAUTION

To immediately stop the robot, release the deadman switch on the back of the T/B. The servo power will turn OFF, and the robot will stop.

The robot will also stop if the [EMG.STOP] switch (emergency stop switch) on the front of the T/B is pressed.



CAUTION

To check whether the robot arm has deviated from its origin position, move the robot arm so that the ABS marks line up. Then check the displayed joint coordinates of the position.

For information on the relationship between the positions of the ABS marks and joint coordinates, refer to "ABS" in the RT ToolBox3 / RT ToolBox3 mini User's Manual.

2.3.1 Installing the teaching pendant (T/B)

Installing the T/B, with turning off the controller power.



CAUTION

Please do not pull the cable of T/B strongly or do not bend it too much. It becomes the breaking of a wire of the cable and the cause of breakage of the connector. Please installing so that stress does not start the cable with the connector itself.

Explain the installation method of T/B below.

- 1) Check that the controller's power supply is OFF.
- 2) Refer to [Fig. 2-7](#) and connect T/B connector to the robot controller.
Use as the upper surface the lock lever, and push in until there is sound.
- 3) Pressing down the lock lever to lock the connector as shown below.

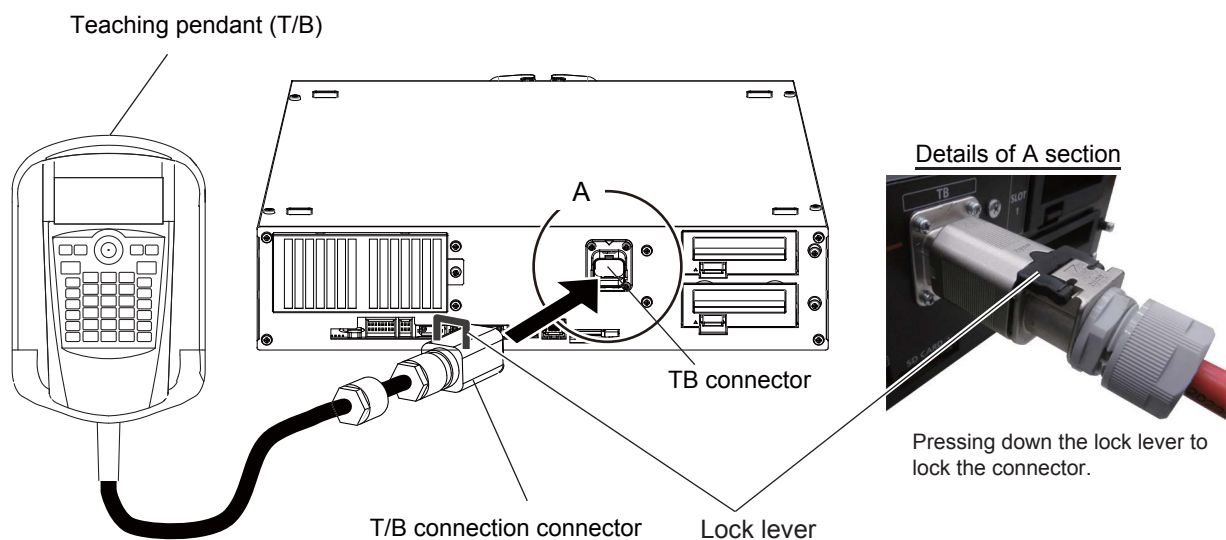


Fig. 2-7: Installing the T/B

2.3.2 Turning ON the control power



CAUTION

Confirm that there are no operators near the robot before turning the power ON.

- 1) Turn the controller power switch ON.
Turns ON the switch of the earth leakage breaker of installation outside.

* Controller software version C2d or later

If the hand conditions (weight, size, and center of gravity) have not been set, error C0330 (Undefined hand condition) will occur upon power on.

Clear the error from the teaching pendant or RT ToolBox3 until the hand conditions are set.

Set the hand conditions using the parameter HNDDAT* (* = 0 to 8).

For further information, refer to "Movement parameter" in the "Instruction Manual/Detailed explanations of functions and operations".

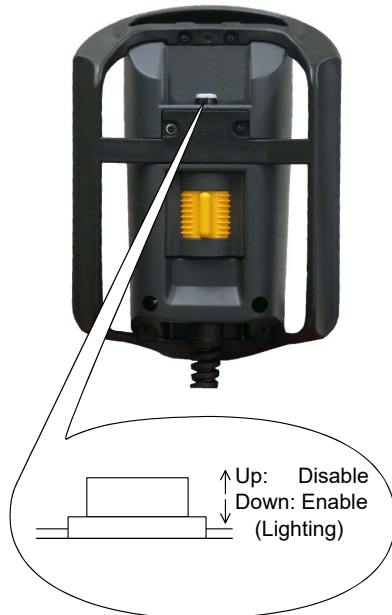
2.3.3 Preparing the T/B

Next, prepare to use the T/B

1) Set the mode of the controller to "MANUAL".

2) Set the T/B [ENABLE] switch to "ENABLE". The menu selection screen will appear.

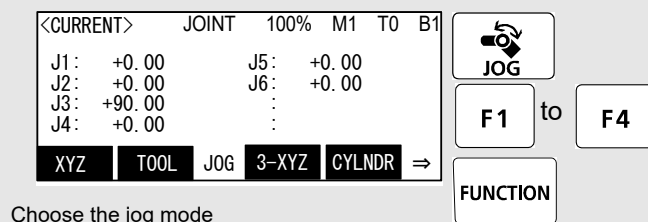
The following operations are carried out with the T/B.



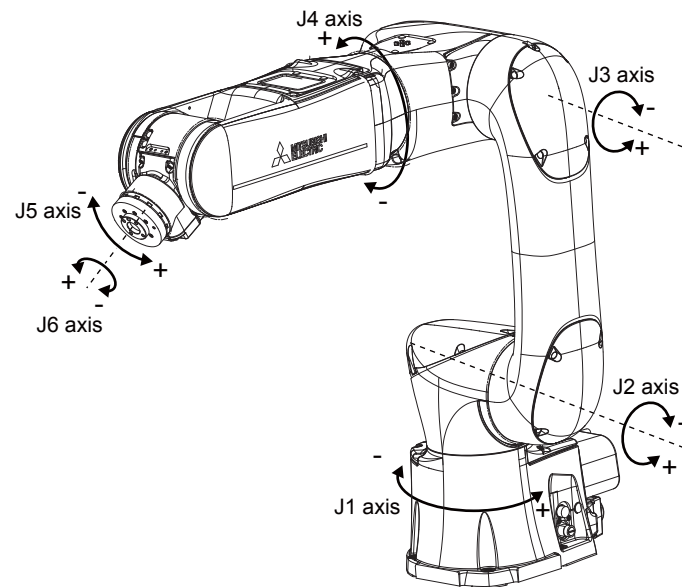
Operating from the T/B

Always set the mode of the controller to "MANUAL", and then set the T/B [ENABLE] switch to "ENABLE". When the T/B is valid, only operations from the T/B are possible. Operations from the external signals will not be accepted. The stop-related operations such as an emergency stop can be performed regardless of the valid/invalid setting.

How to choose the jog mode

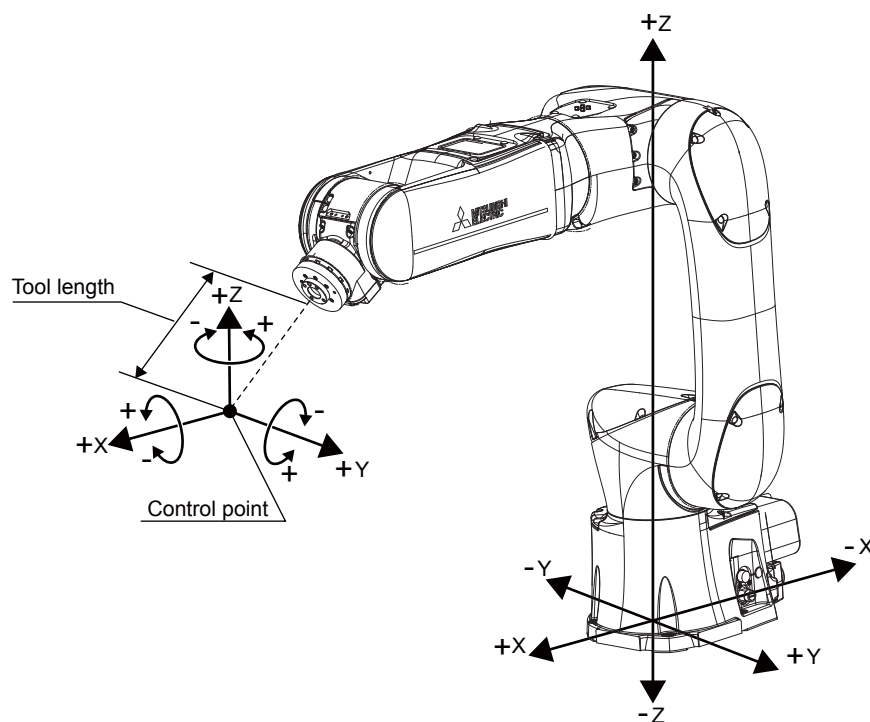


Press the [JOG] key, the jog screen will be displayed, and display the jog mode which can be chosen at the bottom of the screen. Because these correspond to the function key of [F1] - [F4], press the function key corresponding to the jog mode to wish. And, if the [FUNCTION] key is pressed, selection in jog modes other than the present display is possible. The override (100%), the mechanism number (M1), and the tool number (T0), and the base coordinate number (B1) are displayed on the upside of the screen following the present jog mode (JOINT).



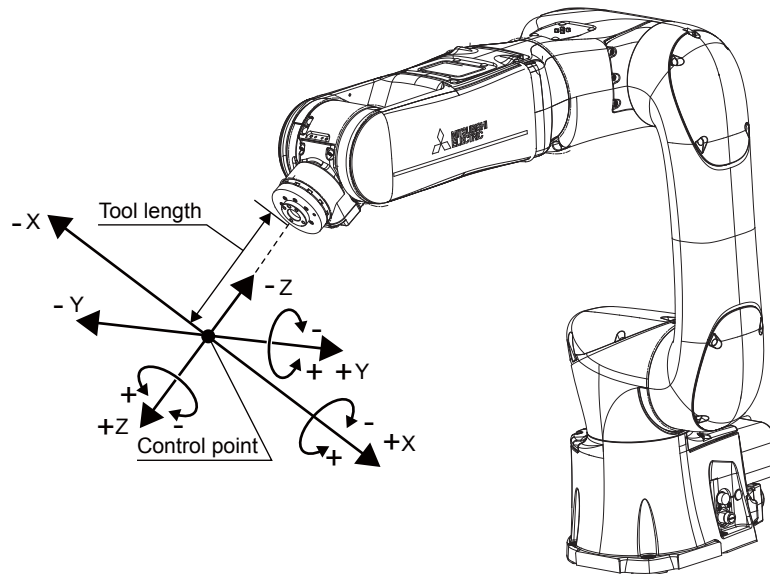
* Each axis moves independently.

Fig. 2-8: JOINT jog operation



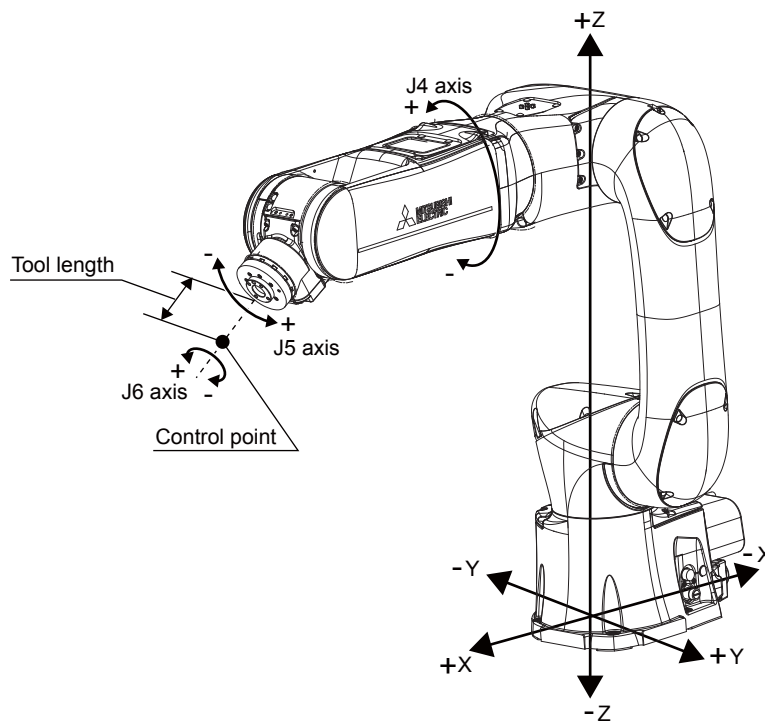
* The axis moves straight along the XYZ coordinate system. The flange surface posture is maintained.
Also, while maintaining the flange surface position, the flange surface posture changes.

Fig. 2-9: XYZ jog operation



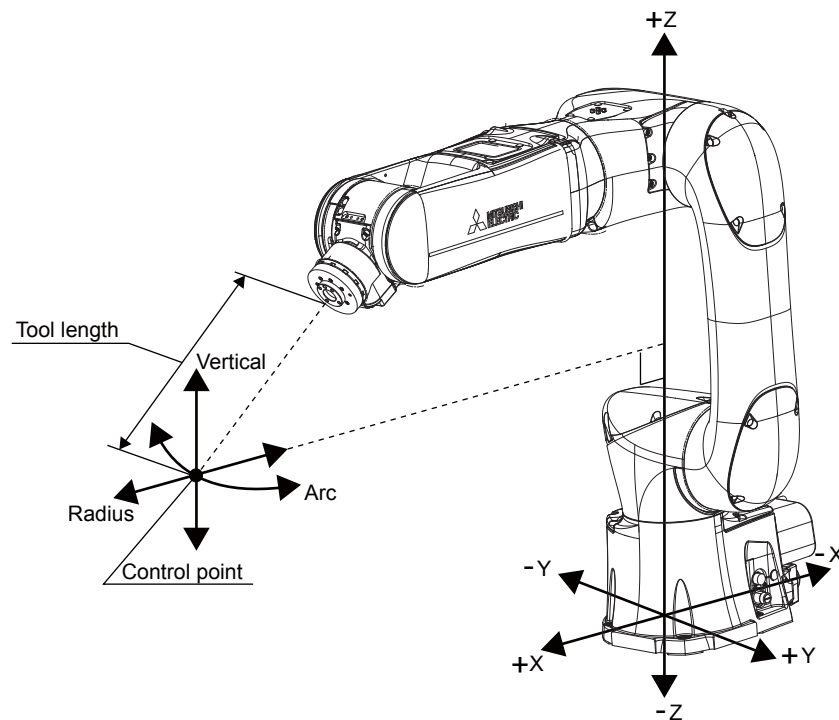
* While maintaining the flange surface posture, the axis moves straight along the tool coordinate system. Also, while maintaining the flange surface position, the flange surface posture changes.

Fig. 2-10: TOOL jog operation



* The axis moves straight along the base coordinate system. At this time, the flange surface posture is not maintained. Also, the flange surface posture changes. The flange surface position does not change at this time.

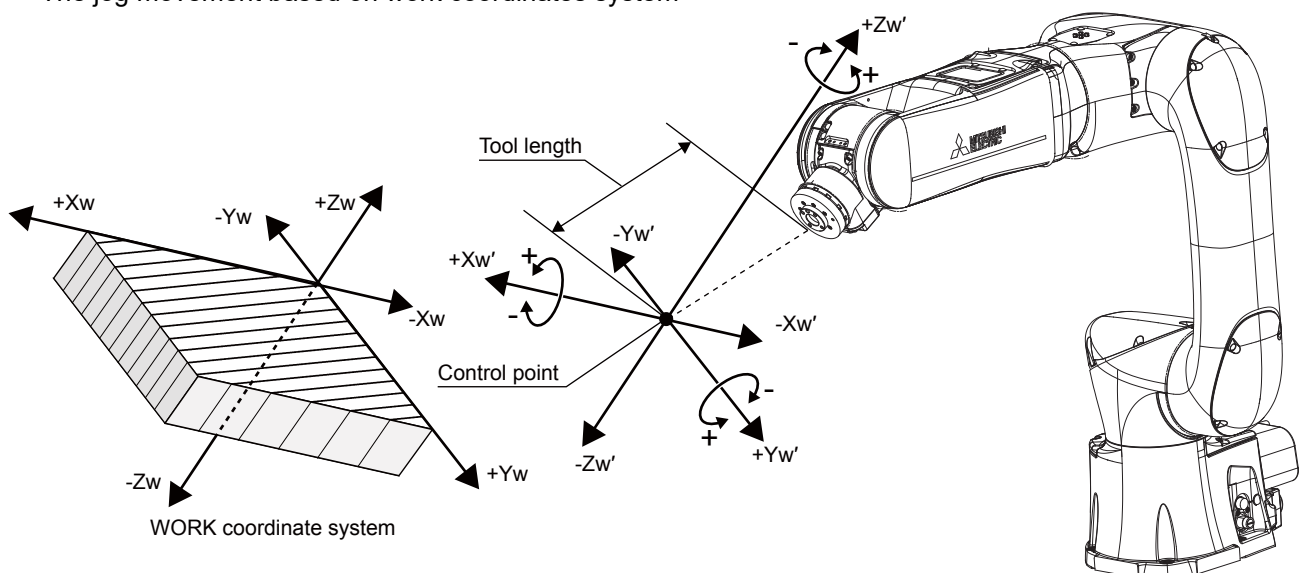
Fig. 2-11: 3-axis XYZ jog operation



* The current position is set as the arc centering on the Z axis, and the axis moves along that arc, expands and contracts in the radius direction, and moves vertically. At this time, the flange surface posture is maintained. Also, while maintaining the flange surface position, the flange surface posture changes.

Fig. 2-12: CYLINDER jog operation

The jog movement based on work coordinates system



* While maintaining the flange surface posture, the axis moves straight along the work coordinate system.

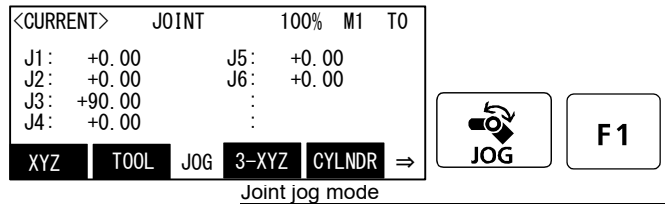
Also, while maintaining the flange surface position, the flange surface posture changes.

* Jog operation around the work coordinate system is available (Ex-T jog). In this jog operation, when the jog operation is performed for the posture elements, the posture rotates on the Xw axis, Yw axis, or Zw axis of the work coordinate system while the control point is changed.

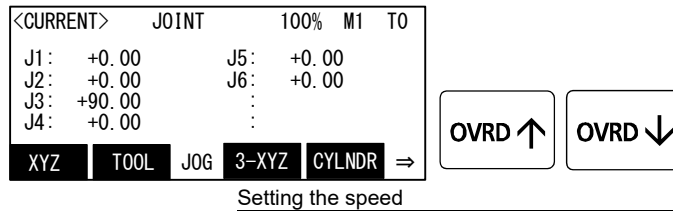
Fig. 2-13: WORK jog operation

(1) JOINT jog operation

Select joint jog mode



Set jog speed



[JOG] Press the key and display the jog screen. ("JOG" is displayed on the screen bottom)

Check that the "joint" in jog mode is displayed on the screen.

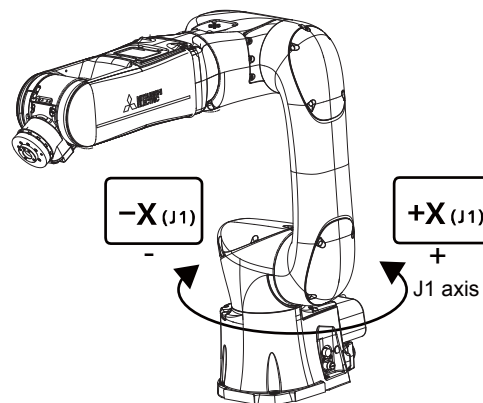
If other jog modes are displayed, please press the function key corresponding to the "joint." (If the jog mode which he wishes under the screen is not displayed, it is displayed that the [FUNCTION] key is pressed)

If it finishes jog operation, press the [JOG] key again, or function key which correspond to "close."

Whenever it presses the key of [OVRD↑], the override goes up. Conversely, if the [OVRD↓] key is pressed, it will go down.

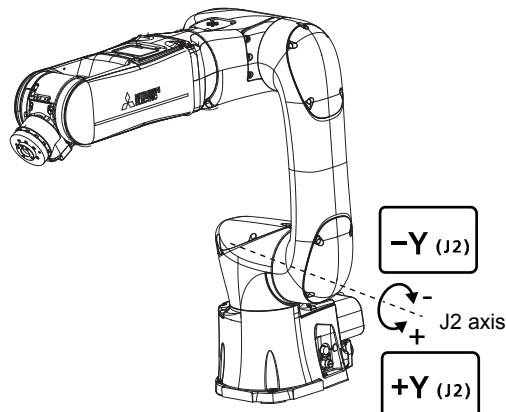
The current setting speed is displayed on screen upper right. Set the override to 10% here for confirmation work.

J1 axis jog operation



- When the [+X (J1)] keys is pressed, the J1 axis will rotate in the plus direction.
When the [-X (J1)] keys is pressed, rotate in the minus direction.

J2 axis jog operation

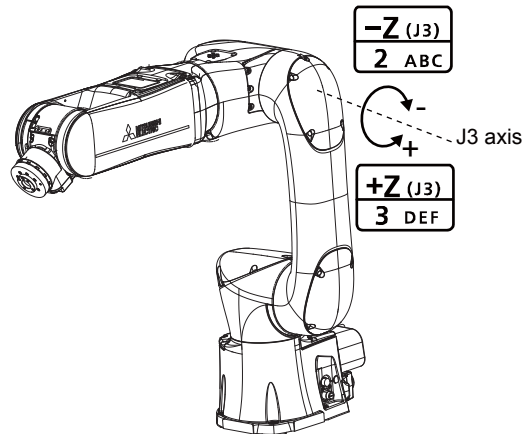


- When the [+Y (J2)] keys is pressed, the J2 axis will rotate in the plus direction.
When the [-Y (J2)] keys is pressed, rotate in the minus direction.

When the robot is in the transportation posture

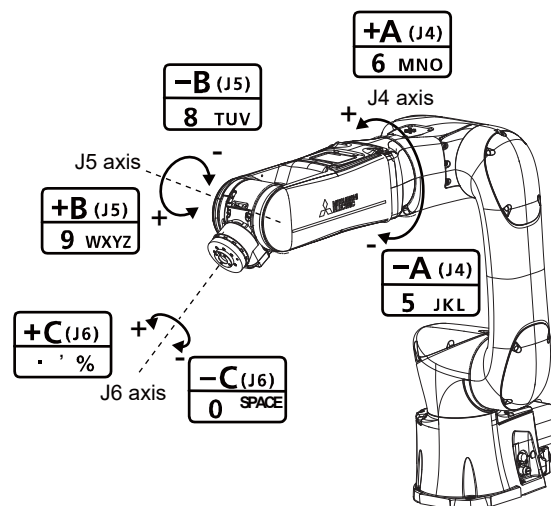
The axes may be outside the movement area. Move these axes toward the inner side of the movement area.

J3 axis jog operation



- When the [+Z (J3)] keys is pressed, the J3 axis will rotate in the plus direction.
When the [-Z (J3)] keys is pressed, rotate in the minus direction.

J4, J5 and J6 axis jog operation



- When the [+A (J4)] key is pressed, the J4 axis will rotate in the plus direction.
When the [-A (J4)] keys is pressed, rotate in the minus direction.
- When the [+B (J5)] keys is pressed, the J5 axis will rotate in the plus direction
When the [-B (J5)] keys is pressed, rotate in the minus direction.
- When the [+C (J6)] keys is pressed, the J6 axis will rotate in the plus direction
When the [-C (J6)] keys is pressed, rotate in the minus direction.

if the buzzer of T/B sounds and the robot does not move

If it is going to move the robot across the operation range, the buzzer of T/B sounds and the robot does not move. In this case, please move to the counter direction.

(2) XYZ jog operation

Select XYZ jog mode

<CURRENT> XYZ 100% P5		 JOG FUNCTION F1 to F4
X: +977.45	A: -180.00	
Y: +0.00	B: +89.85	
Z: +928.24	C: +180.00	
L1: 7	L2: 0	
JOINT TOOL JOG 3-XYZ CYLNR ⇒		

XYZ jog mode

Set jog speed

<CURRENT> XYZ 70% P5		 OVRD ↑	 OVRD ↓
X: +977.45	A: -180.00		
Y: +0.00	B: +89.85		
Z: +928.24	C: +180.00		
L1: 7	L2: 0		
JOINT TOOL JOG 3-XYZ CYLNR ⇒			

Setting the speed

[JOG] Press the key and display the jog screen. ("JOG" is displayed on the screen bottom)

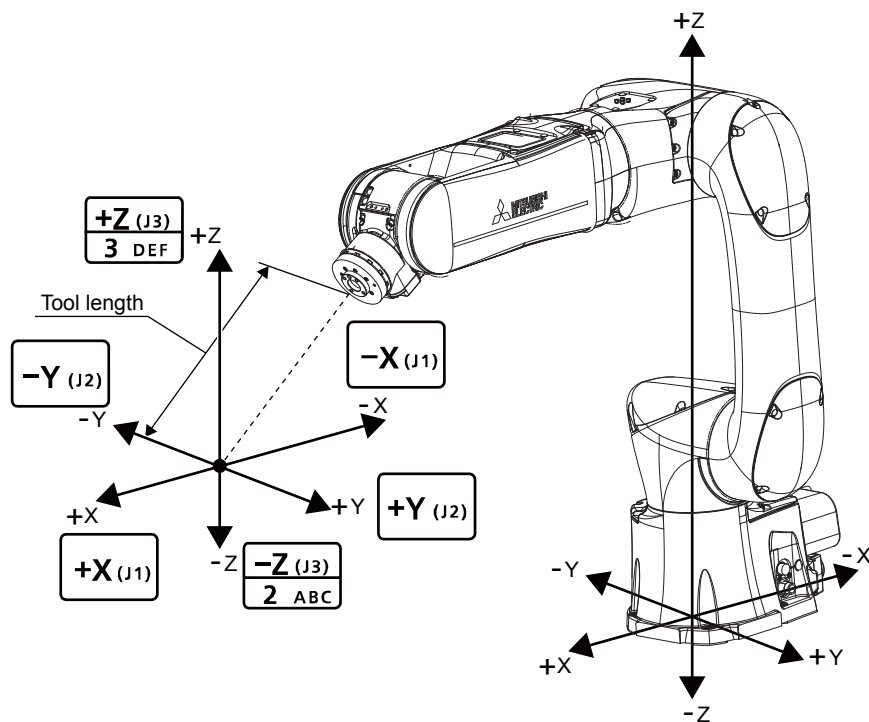
Check that the "XYZ" in jog mode is displayed on the screen.

If other jog modes are displayed, please press the function key corresponding to the "XYZ." (If the jog mode which he wishes under the screen is not displayed, it is displayed that the [FUNCTION] key is pressed)

If it finishes jog operation, press the [JOG] key again, or function key which correspond to "close."

Whenever it presses the key of [OVRD↑], the override goes up. Conversely, if the [OVRD↓] key is pressed, it will go down.

The current setting speed is displayed on screen upper right. Set the override to 10% here for confirmation work.

Moving along the base coordinate system

* The flange direction does not change.

- When the [+X (J1)] keys is pressed, the robot will move along the X axis plus direction.
When the [-X (J1)] keys is pressed, move along the minus direction.
- When the [+Y (J2)] keys is pressed, the robot will move along the Y axis plus direction.
When the [-Y (J2)] keys is pressed, move along the minus direction.
- When the [+Z (J3)] keys is pressed, the robot will move along the Z axis plus direction.
When the [-Z (J3)] keys is pressed, move along the minus direction.

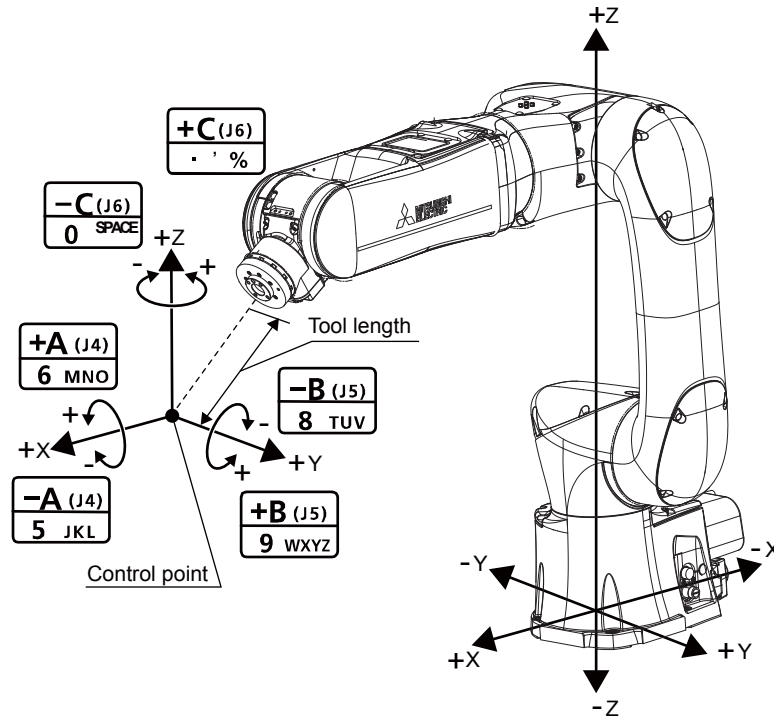
When the robot is in the transportation posture

There are directions from which linear movement is not possible from the transportation posture. In this case, the robot will not move. Refer to section [Page 20, "\(1\) JOINT jog operation"](#), and move the robot to a position where linear movement is possible, and then carry out XYZ jog.

If the buzzer of T/B sounds and the robot does not move

If it is going to move the robot across the operation range, the buzzer of T/B sounds and the robot does not move. In this case, please move to the counter direction.

Changing the flange surface posture



* The control point does not change.

- When the [+A (J4)] key is pressed, the axis will rotate around the X axis in the plus direction. When the [-A (J4)] keys is pressed, rotate in the minus direction.
- When the [+B (J5)] key is pressed, the axis will rotate around the Y axis in the plus direction. When the [-B (J5)] keys is pressed, rotate in the minus direction.
- When the [+C (J6)] key is pressed, the axis will rotate around the Z axis in the plus direction. When the [-C (J6)] keys is pressed, rotate in the minus direction.

Tool length

The default tool length is 0mm, and the control point is the center of the end axis.

After installing the hand, set the correct tool length in the parameters. Refer to the separate manual "Detailed Explanation of Functions and Operations" for details.

(3) TOOL jog operation

Select TOOL jog mode

<CURRENT>	TOOL	100%	P5
X:	+977.45	A:	-180.00
Y:	+0.00	B:	+89.85
Z:	+928.24	C:	+180.00
L1:		L2:	
FL1:	7	FL2:	0
JOINT XYZ JOG 3-XYZ CYLNR =>			
TOOL jog mode			



FUNCTION

F1

F4

Set jog speed

<CURRENT>	TOOL	70%	P5
X:	+977.45	A:	-180.00
Y:	+0.00	B:	+89.85
Z:	+928.24	C:	+180.00
L1:		L2:	
FL1:	7	FL2:	0
JOINT XYZ JOG 3-XYZ CYLNR =>			
Setting the speed			



[JOG] Press the key and display the jog screen. ("JOG" is displayed on the screen bottom)

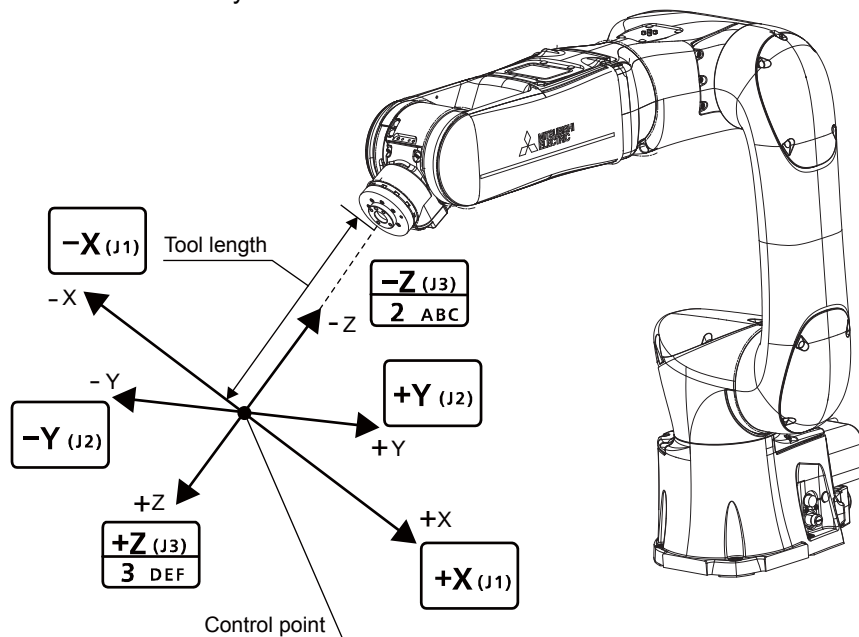
Check that the "TOOL" in jog mode is displayed on the screen.

If other jog modes are displayed, please press the function key corresponding to the "TOOL." (If the jog mode which he wishes under the screen is not displayed, it is displayed that the [FUNCTION] key is pressed)

If it finishes jog operation, press the [JOG] key again, or function key which correspond to "close."

Whenever it presses the key of [OVRD↑], the override goes up. Conversely, if the [OVRD↓] key is pressed, it will go down.

The current setting speed is displayed on screen upper right. Set the override to 10% here for confirmation work.

Moving along the tool coordinate system

* The direction of the flange will not change

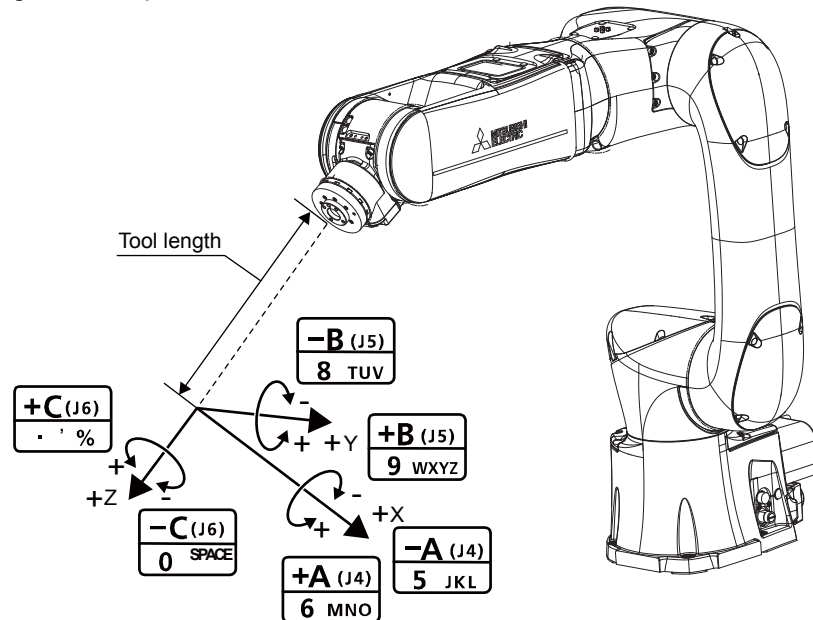
- When the [+X (J1)] keys is pressed, the axis will move along the X axis plus direction of the tool coordinate system.
When the [-X (J1)] keys is pressed, the axis will move along the minus direction.
- When the [+Y (J2)] key is pressed, the axis will move along the Y axis plus direction of the tool coordinate system.
When the [-Y (J2)] key is pressed, the axis will move along the minus direction.
- When the [+Z (J3)] keys is pressed, the axis will move along the Z axis plus direction of the tool coordinate system.
When the [-Z (J3)] keys is pressed, the axis will move along the minus direction.

When the robot is in the transportation posture

There are directions from which linear movement is not possible from the transportation posture. In this case, the robot will not move. Refer to section [Page 20, "\(1\) JOINT jog operation"](#), and move the robot to a position where linear movement is possible, and then carry out XYZ jog.

If the buzzer of T/B sounds and the robot does not move

If it is going to move the robot across the operation range, the buzzer of T/B sounds and the robot does not move. In this case, please move to the counter direction.

Changing the flange surface posture

* The control point does not change.

- When the[+A (J4)] keys is pressed, the X axis will rotate in the plus direction of the tool coordinate system.
When the[-A (J4)] keys is pressed, rotate in the minus direction.
- When the[+B (J5)] keys is pressed, the Y axis will rotate in the plus direction of the tool coordinate system.
When the[-B (J5)] keys is pressed, rotate in the minus direction.
- When the[+C (J6)] keys is pressed, the Z axis will rotate in the plus direction of the tool coordinate system.
When the[-C (J6)] keys is pressed, rotate in the minus direction.

Tool length

The default tool length is 0mm, and the control point is the center of the end axis.

After installing the hand, set the correct tool length in the parameters. Refer to the separate manual "Detailed Explanation of Functions and Operations" for details.

(4) 3-axis XYZ jog operation

Select 3-axis XYZ jog mode

<CURRENT> 3-XYZ 100% P5			
X:	+977.45	A:	-180.00
Y:	+0.00	B:	+89.85
Z:	+928.24	C:	+180.00
L1:		L2:	
FL1:	7	FL2:	0

JOINT XYZ JOG TOOL CYLNRD ⇒

3-axis XYZ jog mode



FUNCTION

F1

to

F4

Set jog speed

<CURRENT> 3-XYZ 70% P5			
X:	+977.45	A:	-180.00
Y:	+0.00	B:	+89.85
Z:	+928.24	C:	+180.00
L1:		L2:	
FL1:	7	FL2:	0

JOINT XYZ JOG TOOL CYLNRD ⇒

Setting the speed



[JOG] Press the key and display the jog screen. ("JOG" is displayed on the screen bottom)

Check that the "3-XYZ" in jog mode is displayed on the screen.

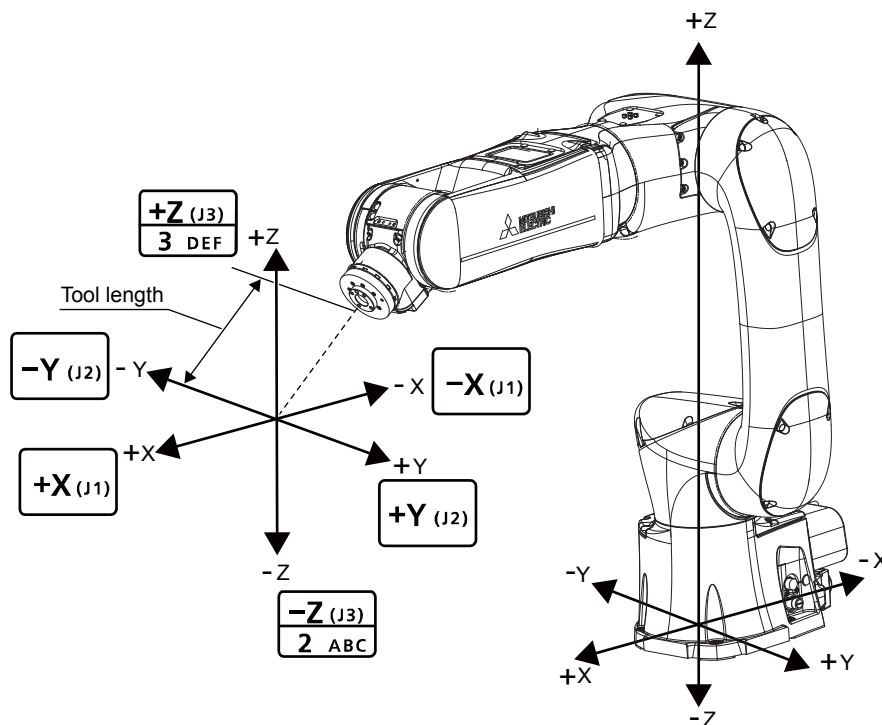
If other jog modes are displayed, please press the function key corresponding to the "3-XYZ" (If the jog mode which he wishes under the screen is not displayed, it is displayed that the [FUNCTION] key is pressed)

If it finishes jog operation, press the [JOG] key again, or function key which correspond to "close."

Whenever it presses the key of [OVRD↑], the override goes up. Conversely, if the [OVRD↓] key is pressed, it will go down.

The current setting speed is displayed on screen upper right. Set the override to 10% here for confirmation work.

Moving along the base coordinate system



* The direction of the flange will change

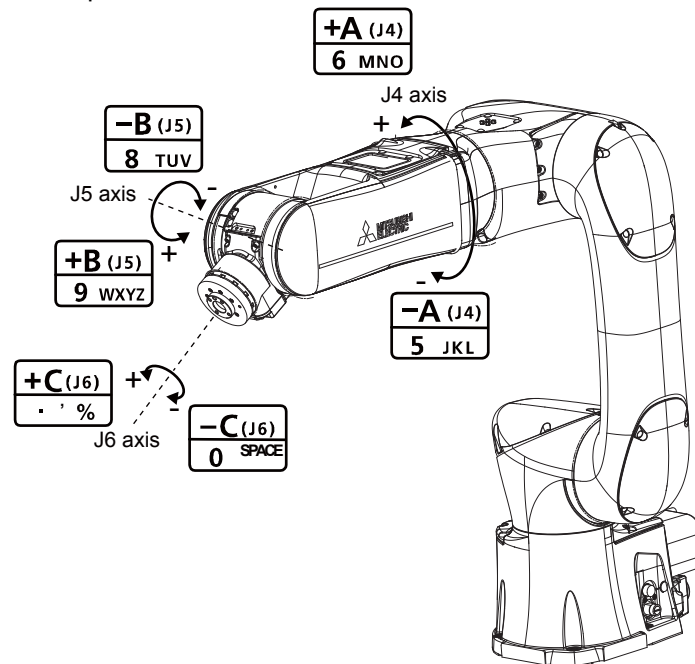
- When the[+X (J1)] keys is pressed, the robot will move along the X axis plus direction.
When the[-X (J1)] keys is pressed, move along the minus direction.
- When the[+Y (J2)] keys is pressed, the robot will move along the Y axis plus direction.
When the[-Y (J2)] keys is pressed, move along the minus direction.
- When the[+Z (J3)] keys is pressed, the robot will move along the Z axis plus direction.
When the[-Z (J3)] keys is pressed, move along the minus direction.

The flange surface end axis posture cannot be maintained with 3-axis XYZ jog.

With 3-axis XYZ jog, the flange surface end axis posture (orientation) is not maintained when moving linearly in the X, Y or Z axis direction.

Use XYZ jog to maintain the posture.

Changing the flange surface posture



* The wrist pose can be changed maintaining the flange's position.

- When the [+A (J4)] key is pressed, the J4 axis will rotate in the plus direction.
At this time, to maintain the flange's position, other axes move simultaneously except J5 and J6.
When the [-A (J4)] key is pressed, the axis will rotate in the minus direction.
- When the [+B (J5)] key is pressed, the J5 axis will rotate in the plus direction.
At this time, to maintain the flange's position, other axes move simultaneously except J4 and J6.
When the [-B (J5)] key is pressed, the axis will rotate in the minus direction.
- When the [+C (J6)] key is pressed, the J6 axis will rotate in the plus direction.
When the [-C (J6)] key is pressed, the axis will rotate in the minus direction.

(5) CYLINDER jog operation

Select cylindrical jog mode

<CURRENT>	CYLINDER	100%	P5
X:	+977.45	A:	-180.00
Y:	+0.00	B:	+89.85
Z:	+928.24	C:	+180.00
L1:		L2:	
FL1:	7	FL2:	0
JOINT	XYZ	JOG	TOOL 3-XYZ ⇒

CYLINDER jog mode



FUNCTION

F1

to F4

Set jog speed

<CURRENT>	CYLINDER	70%	P5
X:	+977.45	A:	-180.00
Y:	+0.00	B:	+89.85
Z:	+928.24	C:	+180.00
L1:		L2:	
FL1:	7	FL2:	0
JOINT	XYZ	JOG	TOOL 3-XYZ ⇒

Setting the speed

OVRD ↑

OVRD ↓

[JOG] Press the key and display the jog screen. ("JOG" is displayed on the screen bottom)

Check that the "CYLINDER" in jog mode is displayed on the screen.

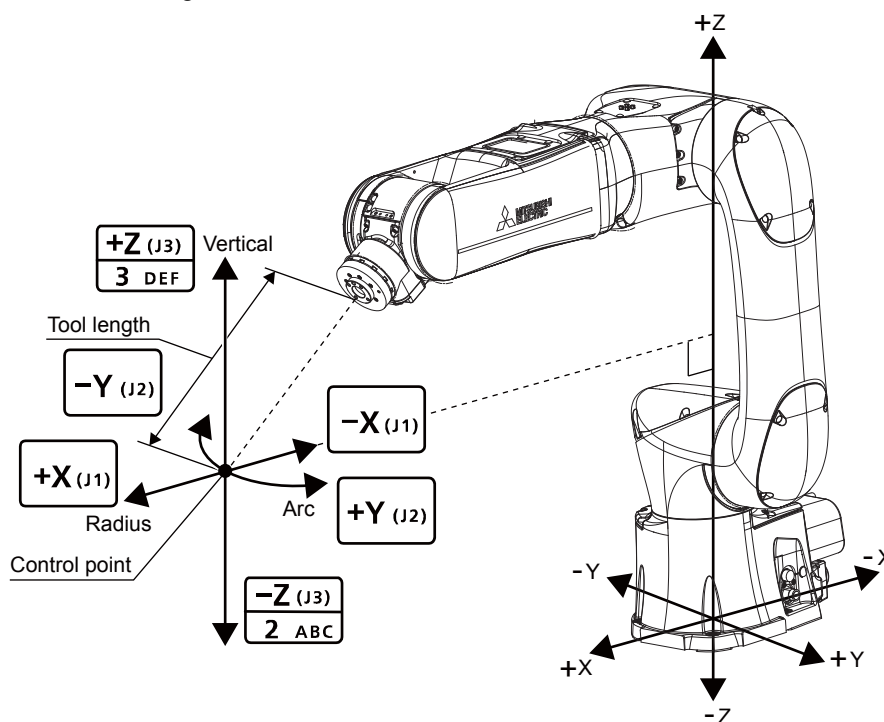
If other jog modes are displayed, please press the function key corresponding to the "CYLINDER." (If the jog mode which he wishes under the screen is not displayed, it is displayed that the [FUNCTION] key is pressed)

If it finishes jog operation, press the [JOG] key again, or function key which correspond to "close."

Whenever it presses the key of [OVRD↑], the override goes up. Conversely, if the [OVRD↓] key is pressed, it will go down.

The current setting speed is displayed on screen upper right. Set the override to 10% here for confirmation work.

Moving along an arc centering on the Z axis

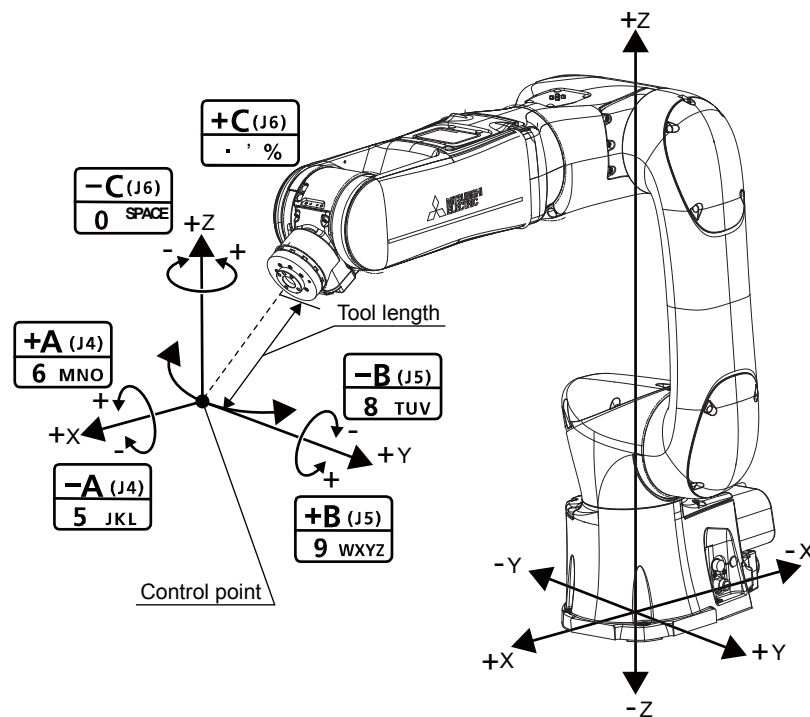


* The direction of the frange will not change.

Assuming that the current position is on an arc centering on the Z axis, the robot moves along that arc.

- When the[+X (J1)] keys is pressed, the robot will expand in the radial direction.
When the[-X (J1)] keys is pressed, contract in the radial direction.
- When the[+Y (J2)] keys is pressed, the robot will move along the arc in the plus direction.
When the[-Y (J2)] keys is pressed, move in the minus direction.
- When the[+Z (J3)] keys is pressed, the robot will move along the Z axis plus direction.
When the[-Z (J3)] keys is pressed, move along the minus direction.

Changing the flange surface posture



* The flange position does not change.
This is the same as the A, B and C axis XYZ jog operation.

- When the [+A (J4)] key is pressed, the axis will rotate around the X axis in the plus direction.
When the [-A (J4)] key is pressed, rotate in the minus direction.
- When the [+B (J5)] key is pressed, the axis will rotate around the Y axis in the plus direction.
When the [-B (J5)] key is pressed, rotate in the minus direction.
- When the [+C (J6)] key is pressed, the axis will rotate around the Z axis in the plus direction.
When the [-C (J6)] key is pressed, rotate in the minus direction.

(6) Work jog operation

Setting of the work coordinates system is necessary.

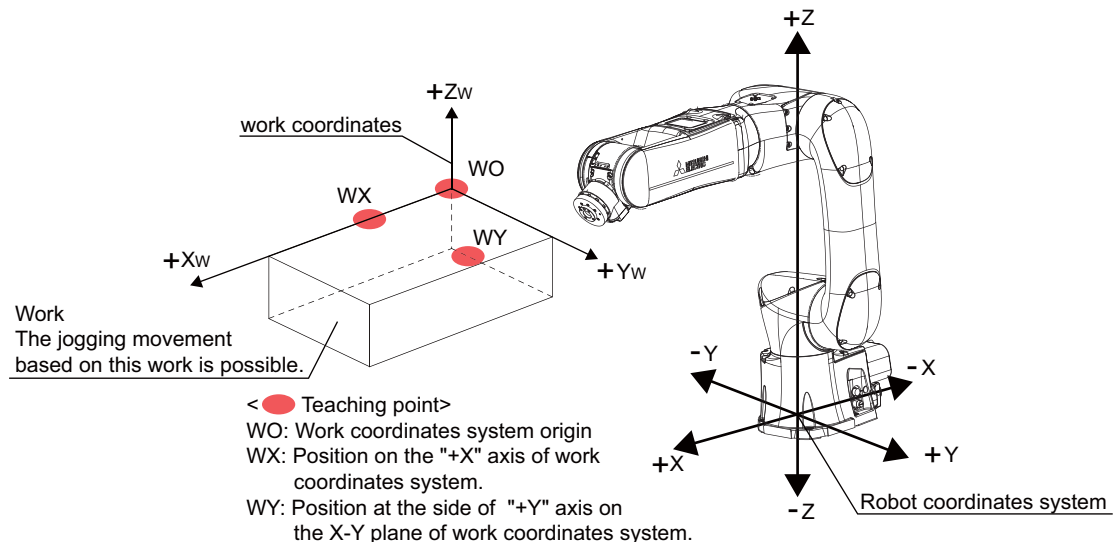
By this jog operation, robot can be move along with the direction of work (or working table etc.), so teaching operations get easier.

When jog operation, select by which work coordinates the robot moves

The setting method of the work coordinates system using T/B (R32TB) is shown in the following.

(Parameter: Setting the coordinate value to WK n CORD ("n" is meaning the number (1-8) of work coordinates) can also set up the work coordinates system. Refer to the separate manual "Detailed Explanation of Functions and Operations" for details of parameter.)

The work coordinates system teaches and sets up the three points (WO, WX, WY).

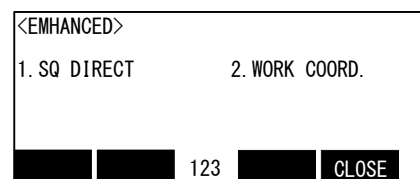
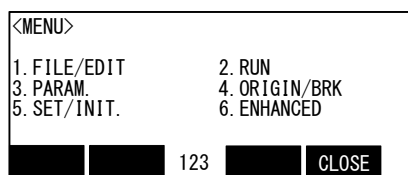


[Supplement] : The coordinate values which use all three teaching points for setting of the work coordinates system are each only X, Y, and the Z-axis. Although the coordinate value of A, B, and C axis is not used, positioning will get easy if the XYZ jog or TOOL jog movement is effected with the same value. (The direction of the hand is the same)

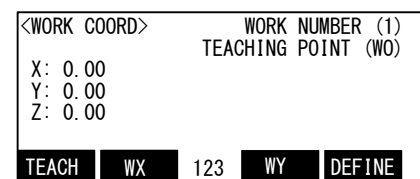
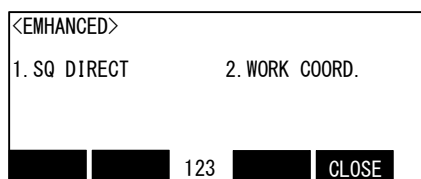
Fig. 2-14: Setting of the work coordinates system (teaching point)

The setting (definition) method of the work coordinates system is shown in the following.

- 1) Select "6.ENHANCED" screen on the <MENU> screen.



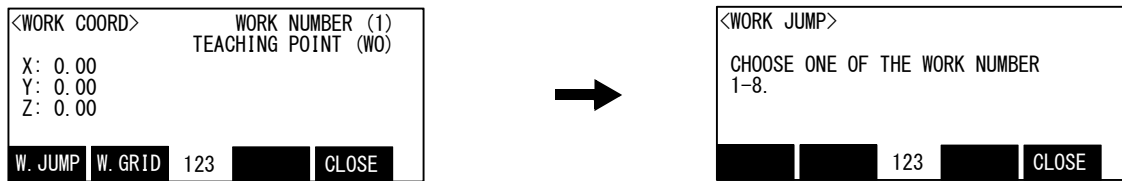
- 2) Press the [2] keys in the menu screen and select "2. WORK COORD."



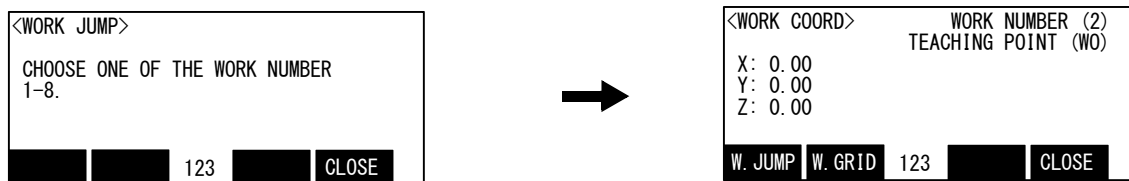
The screen shows the coordinate value of the origin (WO) of the work coordinates number 1.

3) Selection of the work coordinates number

Press the [FUNCTION] keys, and display "W: JUMP" function. Press the function key corresponding to "W: JUMP"



Press numeral key [1] - [8] and specify the work coordinates number. The coordinate value of the specified work coordinates system is displayed.

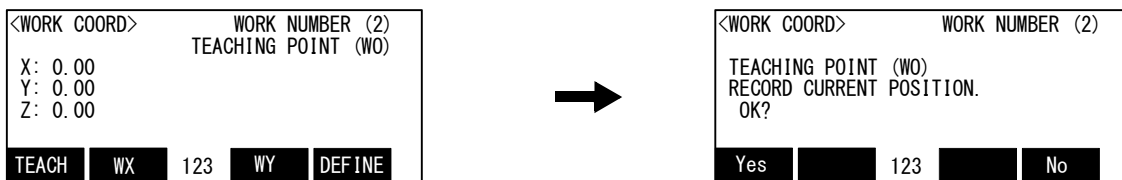


Operation will be canceled if the [CLOSE] key is pressed.

The screen is the example which specified the work coordinates number 2. ("2" at the upper right of the screen)

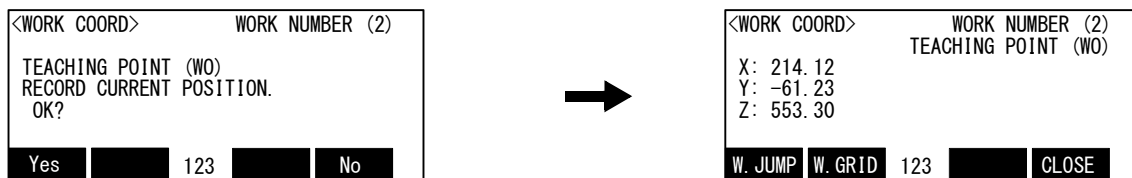
4) The teaching of the work coordinates system

Teach the three points shown in Fig. 2-14. Confirm the name currently displayed on the "TEACHING POINT" at the upper right of the screen. If it differs, press the function key corresponding to each point(WO, WX, WY) to teach. Move the robot's arm by jog operation (other jogging movement), and press the function key corresponding to "TEACH."([F1]) The confirmation screen is displayed.



Specify the teaching point [WO],[WX],[WY]
teaching the position [TEACH]

Presses the function key corresponding to "Yes", the robot's current position is registered, and the registered coordinates value is displayed. Operation will be canceled if the [CLOSE] key is pressed.



Teach the three points, WO, WX, and WY, by the same operation.

The position data taught here is each registered into the following parameters. ("n" means the work coordinates numbers 1-8)

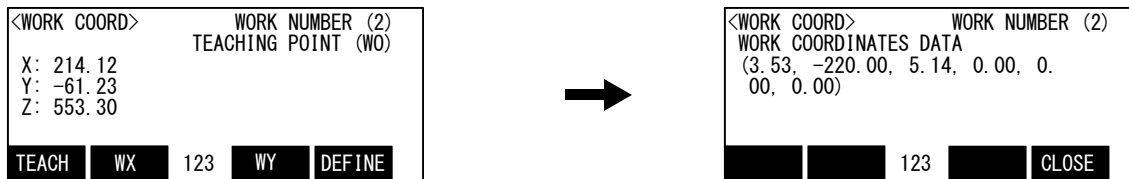
WO= parameter: WKnWO

WX= parameter: WKnWX

WY= parameter: WKnWY

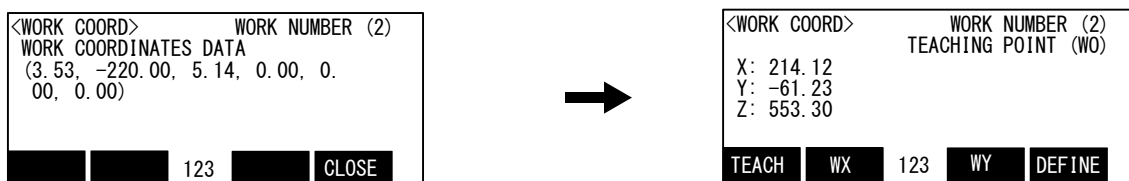
5) Setting of work coordinates (definition)

If the function key corresponding to "DEFINE" ([F1]) is pressed, the work coordinates system will be calculated using the three points, and the result will be displayed.



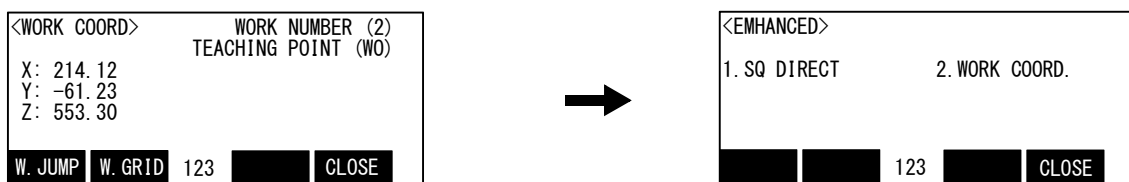
The alarm occurs if the work coordinates system is incalculable. (There are the three points on the straight line, or the two points have overlapped) In this case, reset alarm and re-teach the three points. This work coordinate data is registered into parameter: WKnCORD. ("n" means the work coordinates numbers 1-8)

If the function key corresponding to "CLOSE" is pressed, it will return to the previous screen.

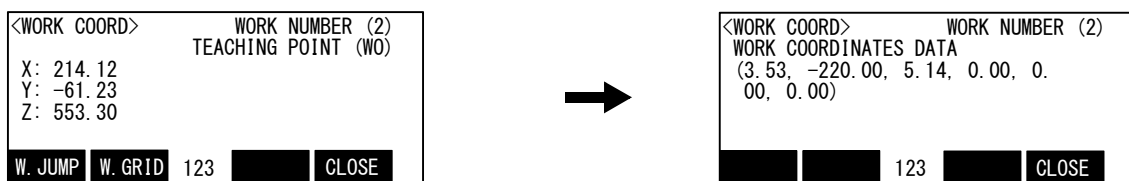


6) Finishing of setting the work coordinates

Press the [FUNCTION] keys, and display "CLOSE" function. Press the function key corresponding to "CLOSE". Returns to the <MENU> screen.

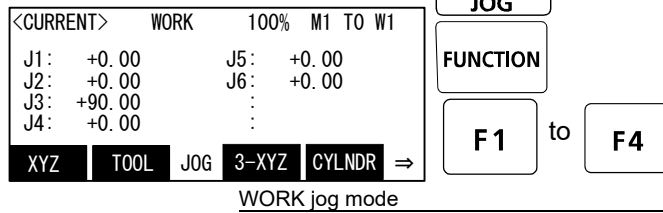


Although setting of work coordinates is finishing above, confirmation of work coordinates can be done by pressing the function key corresponding to "W GRID."([F2])



Return to the previous screen by pressing the [CLOSE] ([F4]) key.

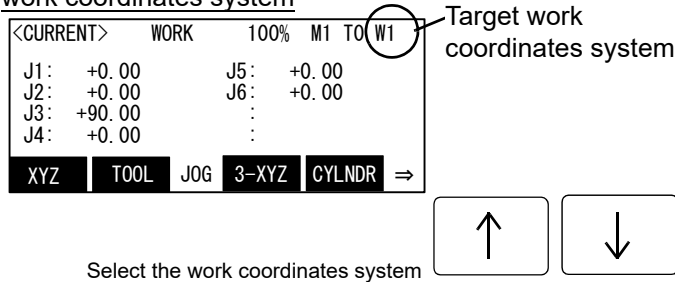
Then, the operation method of the work jog is shown. Change to the work jog after nearing the work.

Select WORK jog mode

[JOG] Press the key and display the jog screen. ("JOG" is displayed on the screen bottom)

Check that the "WORK" in jog mode is displayed on the screen.

If other jog modes are displayed, please press the function key corresponding to the "WORK." (If the jog mode which he wishes under the screen is not displayed, it is displayed that the [FUNCTION] key is pressed) If it finishes jog operation, press the [JOG] key again, or function key which correspond to "close."

Confirmation and selection of the work coordinates system

Confirm the target work coordinates system.

The current target number is displayed on the screen upper right. (W1 - W8)

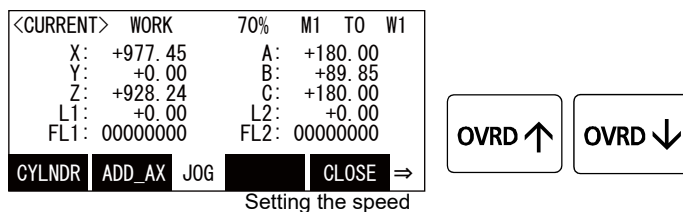
The number of work coordinates can be changed by the arrow key [Upper arrow], [Lower arrow]

Push the key [Upper arrow], the number will increase. (W1, W2, W8) Conversely, push the key [Lower arrow], the number will decrease

CAUTION

Always confirm that the number of the target work coordinates system is displayed correctly (Display of W1-W8 at the upper right of the screen)

If mistaken, the robot will move in the direction which is not meant and will cause the damage and the personal injuries.

Set jog speed

Whenever it presses the key of [OVRD(Upper arrow)], the override goes up. Conversely, if the [OVRD(Lower arrow)] key is pressed, it will go down.

The current setting speed is displayed on screen upper right.

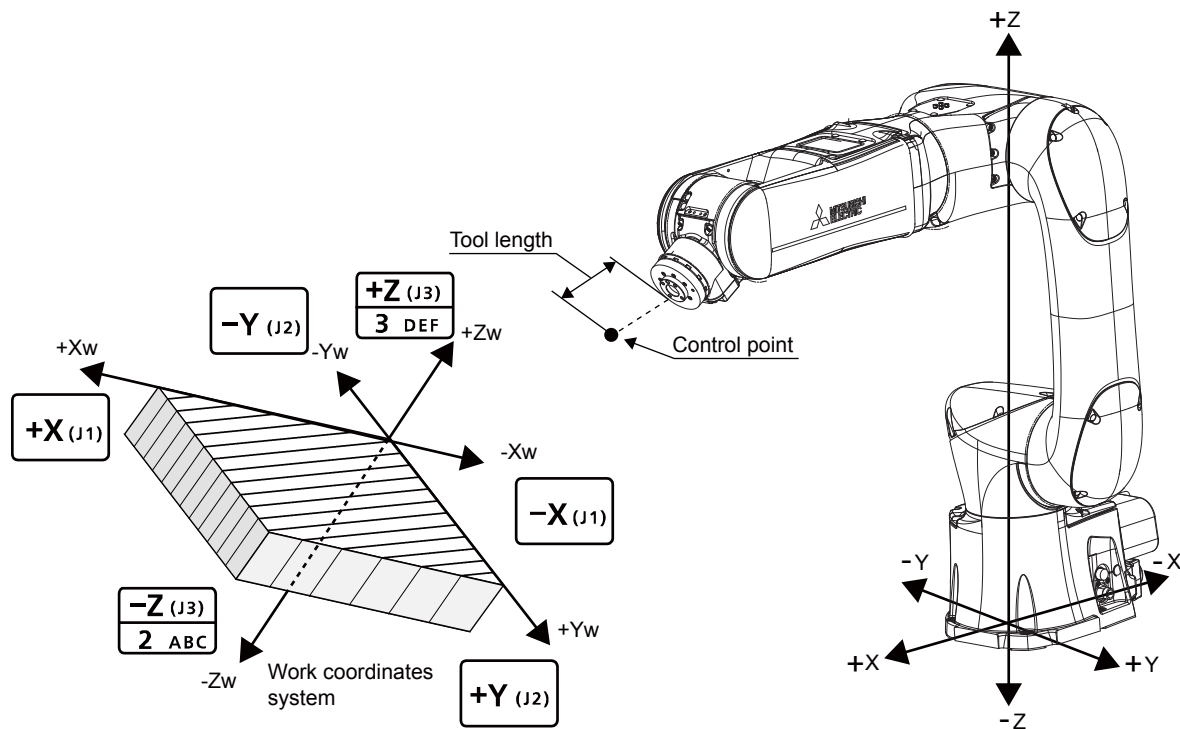
Set the override to 10% here for confirmation work

The WORK jog operation and the Ex-T jog operation can be switched by setting the parameters WK1-JOGMD to WK8JOGMD of each work coordinates system.

The respective operations are as follows.

WORK jog operation mode	Conventional WORK jog	Ex-T jog
Parameters WK _n JOGMD (n is 1 to 8) setting	0 (initial value)	1
XYZ key operation	Moves along each axis of the work coordinates system	Same as the conventional WORK jog
ABC key operation	With the control point position maintained, the direction changes along the work coordinates system.	While the control point position is changed, the direction changes on each axis of the work coordinates system.

The jog movement based on work coordinates system



* The direction of the flange will not change.

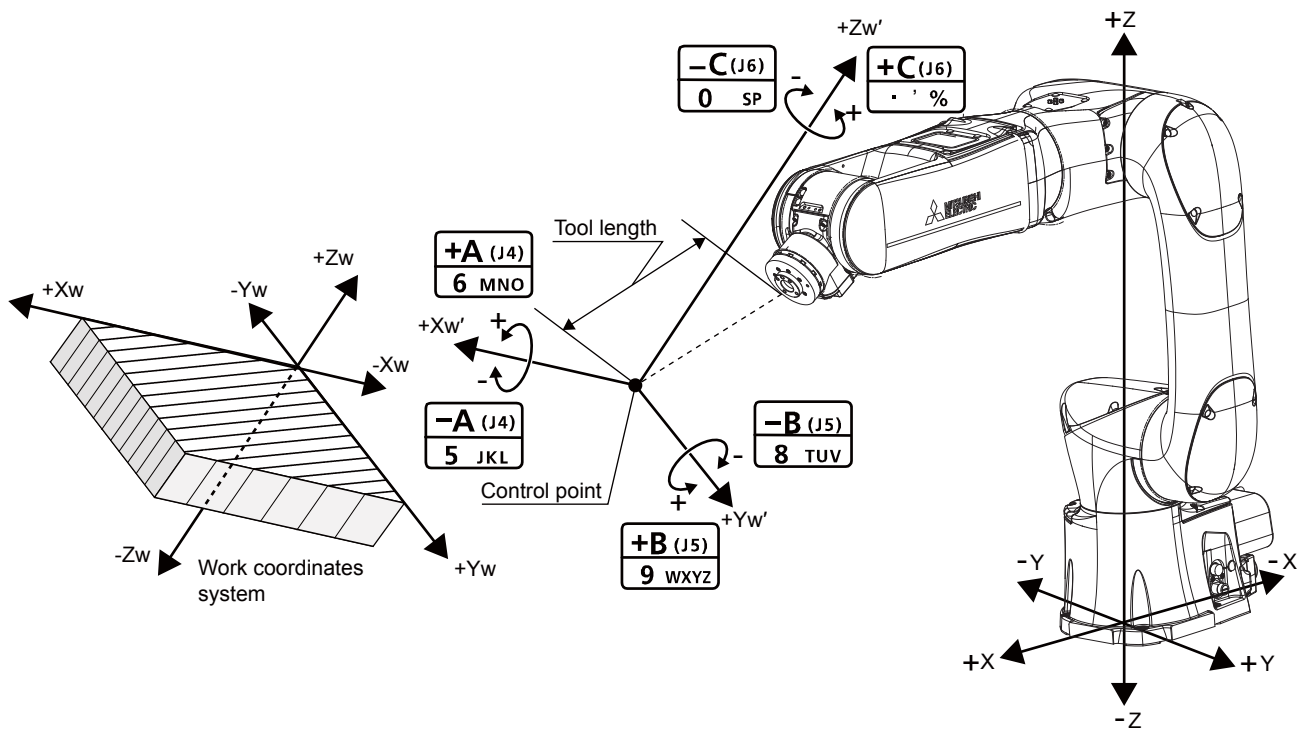
Move the control point with a straight line in accordance with the work coordinates system

- When the [+X (J1)] keys is pressed, the robot will move along the X axis (Xw) plus direction on the work coordinates system.
When the [-X (J1)] keys is pressed, Move along the minus direction.
- When the [+Y (J2)] keys is pressed, the robot will move along the Y axis (Yw) plus direction on the work coordinates system.
When the [-Y (J2)] keys is pressed, Move along the minus direction.
- When the [+Z (J3)] keys is pressed, the robot will move along the Z axis (Zw) plus direction on the work coordinates system.
When the [-Z (J3)] keys is pressed, Move along the minus direction.

When the X, Y, or Z keys is used, the operation is the same in the WORK jog and the Ex-T jog modes.

Changing the flange surface posture

<1> Work jog mode

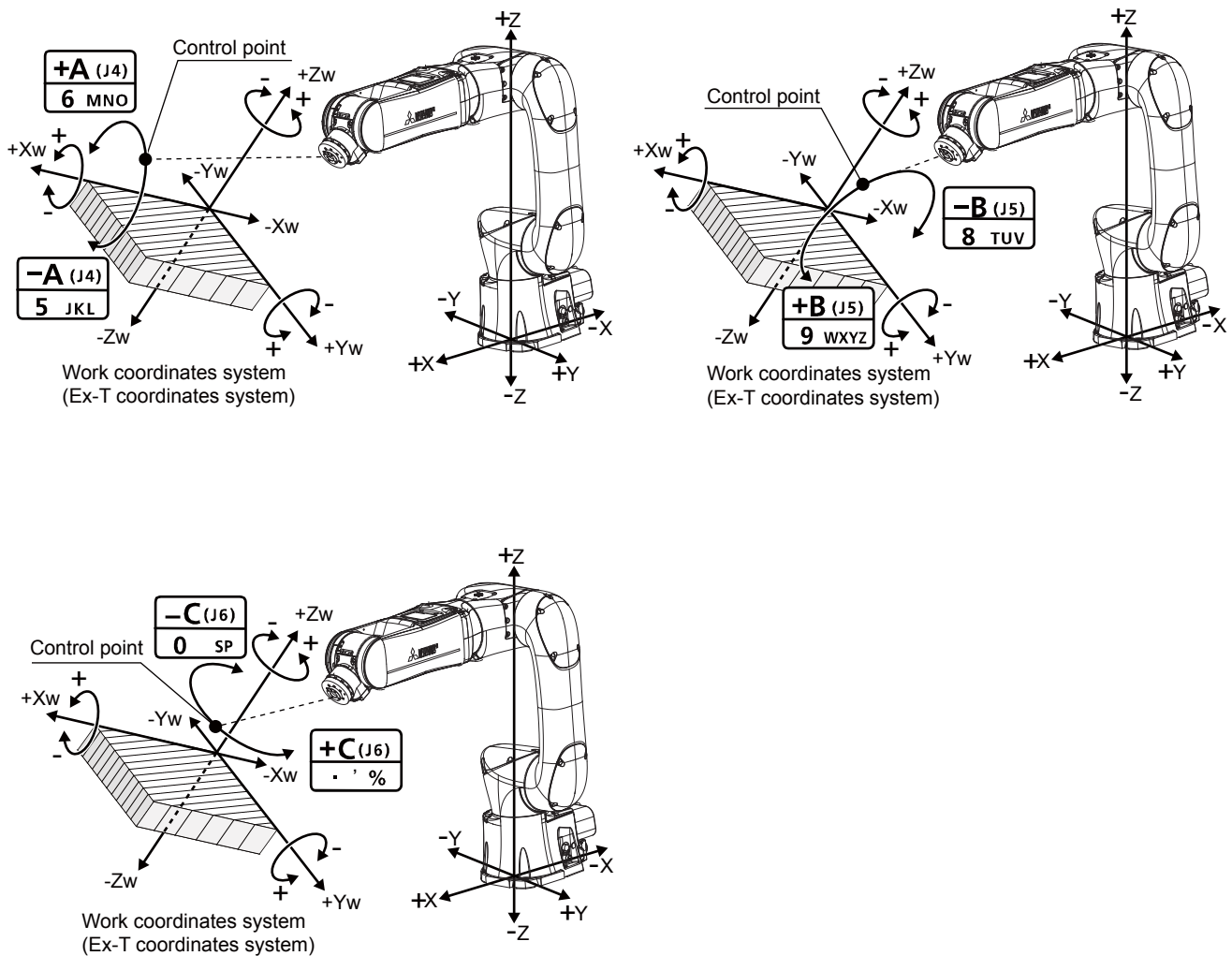


* The position of the control point does not change.

Change the direction of the flange in accordance with the work coordinates system.

- When the [+A (J4)] key is pressed, the axis will rotate around the X axis in the plus direction.
When the [-A (J4)] keys is pressed, rotate in the minus direction.
- When the [+B (J5)] key is pressed, the axis will rotate around the Y axis in the plus direction.
When the [-B (J5)] keys is pressed, rotate in the minus direction.
- When the [+C (J6)] key is pressed, the axis will rotate around the Z axis in the plus direction.
When the [-C (J6)] keys is pressed, rotate in the minus direction.

<2> Ex-T jog mode



* The control point rotates around each axes of work coordinates system (Ex-T coordinates system). When the $[+A (J4)]$ or the $[-A (J4)]$ key is pressed, the control point rotates around the X_w axis. When the $[+B (J5)]$ or the $[-B (J5)]$ key is pressed, the control point rotates around the Y_w axis. When the $[+C (J6)]$ or the $[-C (J6)]$ key is pressed, the control point rotates around the Z_w axis.

- When the $[+A (J4)]$ keys is pressed, the control point will rotate in the plus direction around the X axis (X_w) of work coordinates system (Ex-T coordinates system).
When the $[-A (J4)]$ keys is pressed, the control point will rotate in the minus direction.
- When the $[+B (J5)]$ keys is pressed, the control point will rotate in the plus direction around the Y axis (Y_w) of work coordinates system. (Ex-T coordinates system).
When the $[-B (J5)]$ keys is pressed, the control point will rotate in the minus direction.
- When the $[+C (J6)]$ keys is pressed, the control point will rotate in the plus direction around the Z axis (Z_w) of work coordinates system. (Ex-T coordinates system).
When the $[-C (J6)]$ keys is pressed, the control point will rotate in the minus direction.

When the robot is in the transportation posture

There are directions from which linear movement is not possible from the transportation posture. In this case, the robot will not move. Refer to section [Page 20, "\(1\) JOINT jog operation"](#), and move the robot to a position where linear movement is possible, and then carry out XYZ jog.

If the buzzer of T/B sounds and the robot does not move

If it is going to move the robot across the operation range, the buzzer of T/B sounds and the robot does not move. In this case, please move to the counter direction.

Tool length

The default tool length is 0mm, and the control point is the center of the end axis.
After installing the hand, set the correct tool length in the parameters. Refer to the separate manual "Detailed Explanation of Functions and Operations" for details.

2.3.4 Setting the hand parameters

Set the parameters to set the hand I/O type and the hand condition according to the robot hand to be used. Refer to the separate volume, "Instruction Manual/Detailed Explanations of Functions and Operations" for how to set parameters.

Table 2-4:Hand parameter

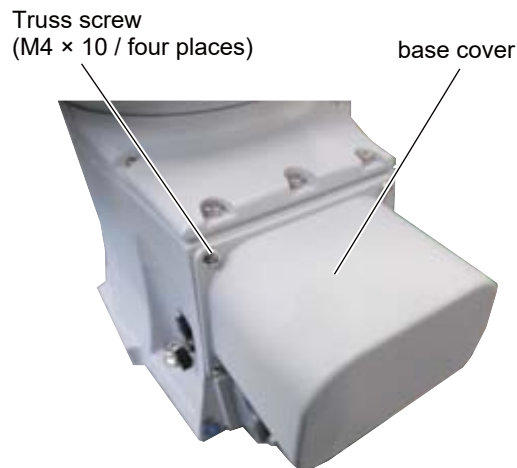
Parameter	Parameter name	Details explanation	Factory setting
Hand I/O type	HIOTYPE	Set either the sink type or the source type for the solenoid valve and the logic of the hand input signal. -1: Not set 0: Source type 1: Sink type	-1
Hand condition	HNDDAT* * is 0 to 8	Set the initial condition of the hand. (Specify with the tool coordinate system.) (Weight, size X, size Y, size Z, center of gravity X, center of gravity Y, center of gravity Z) Unit: Kg, mm  CAUTION Set the hand and workpiece conditions correctly. If a value lower than the actual load is set, the life of robot mechanical parts may be shortened.	The setting varies depending on the model.
Hand condition setting check	HNDCHK	Enable or disable whether to notify that the HNDDAT parameter is not set. 0: Disabled 1: Enabled When this parameter is set to "1" (enabled), error C0330 will occur upon power on of the controller if all the values of the hand condition parameter HNDDAT* (* = 0 to 8) are not set (not changed from the initial values). When this parameter is set to "0" (disabled), the above settings are not checked, and error C0330 does not occur even if the hand conditions are not set. This parameter is available with controller software version C2d or later.	1

3 Installing the option devices

3.1 Installing the solenoid valve set

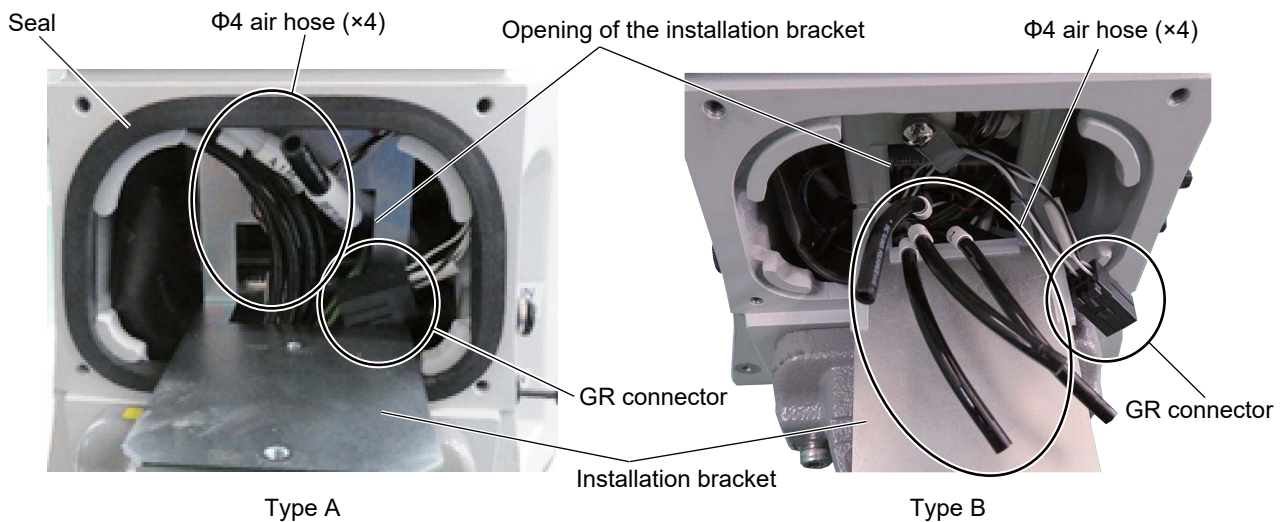
This section explains how to install and connect the solenoid valve set. The installation steps are shown below. Turn off the power of the robot controller before attempting these steps.

- 1) Remove the M4 truss screws attaching the base cover to the back of the robot (four places), then gently remove the base cover. It may be difficult to remove the cover depending on the posture of the robot. Use Jog operation to move the robot into a posture that makes removal of the cover easier.



- 2) Pull out the GR connector and the $\phi 4$ air hoses so that they stick out past the opening of the installation bracket.

There are two types of layouts for the inside of the base: types A and B.



Type A: Products manufactured in or before August 2021

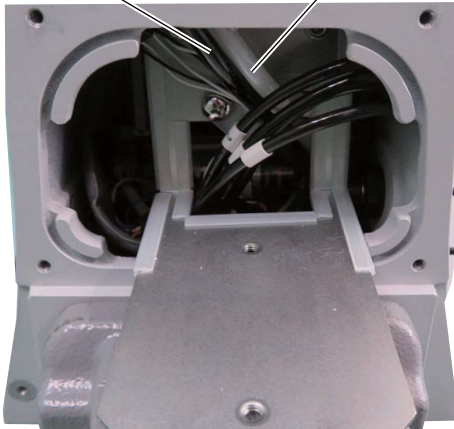
Type B: Products manufactured in or after September 2021

The date of manufacture of the robot can be found in the DATE column on the rating plate located on the robot base.

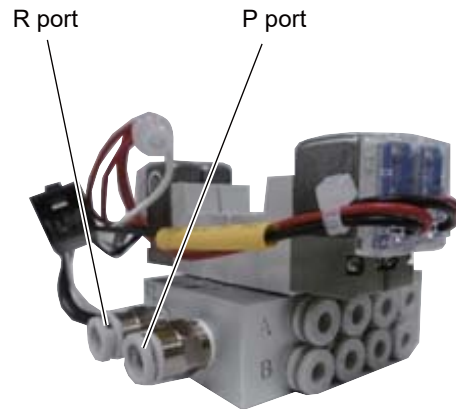
Images of type B are used in the following explanations. The solenoid valve is installed in the same way for both types A and B.

- 3) First, connect the $\phi 6$ RETURN air hose (black) to the R port (quick coupling) of the solenoid valve set. Then, connect the $\phi 6$ AIR IN air hose (white) to the P port (quick coupling) of the solenoid valve set.

$\phi 6$ RETURN air hose (black) $\phi 6$ AIR IN air hose (white)



Robot arm



Solenoid valve set

⚠ CAUTION

When installing the solenoid valve, ensure that the $\phi 6$ air hose (white) is located under the clamp as shown in the image below. Otherwise, the air hose may be bent.

⚠ CAUTION

When connecting the $\phi 6$ air hose (white), ensure that it is not bent at sharp angles. Ensure that the hose fits securely inside the robot.



Clamp

 $\phi 6$ air hose (white)

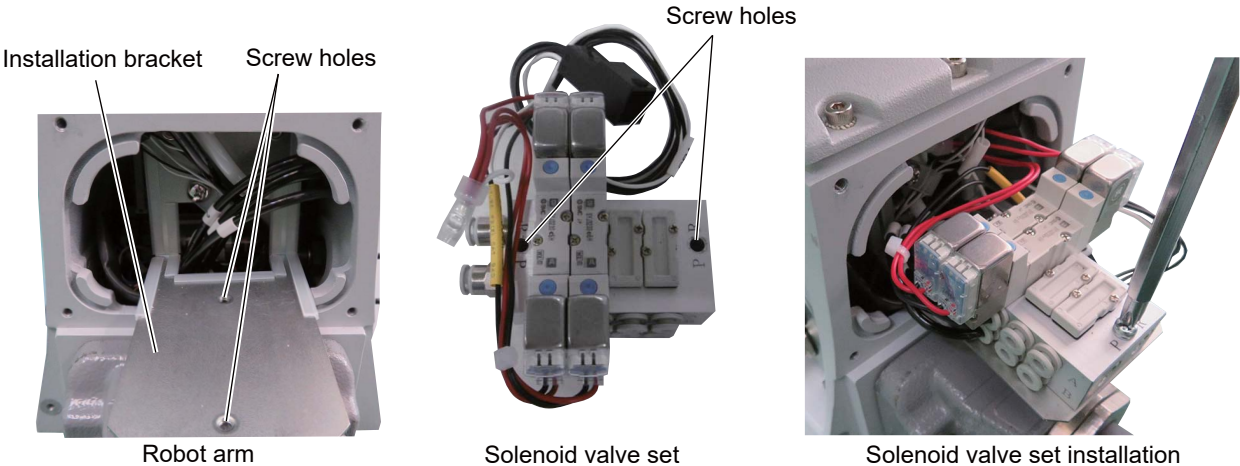
✗ (The air hose is located above the clamp.)

● (The air hose is located under the clamp.)

⚠ CAUTION

When using the robot in oil mist environments, the inside of the robot arm can be pressurized to ensure the robot reliably meets IP54 beyond the warranty period. Instead of connecting the $\phi 6$ RETURN air hose to the solenoid valve, it can be used for supplying dry air to pressurize the robot. For further information on using the robot in oil-mist environments, refer to the section "Protection specifications" in the Standard Specifications Manual.

4) Use the provided M4 × 30 screws (2 pcs.) to secure the solenoid valve set to the bracket on robot.



5) Connect the robot's $\phi 4$ air hoses ($\times 4$) to the A and B ports on the solenoid valve set.
[Table 3-1](#) shows how tools can be connected to the solenoid valves and the operations of each port (open/close).
Refer to [Table 3-1](#) when plumbing solenoid valves.

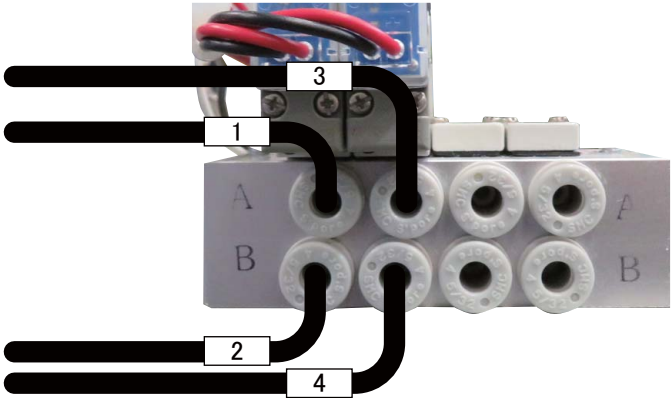
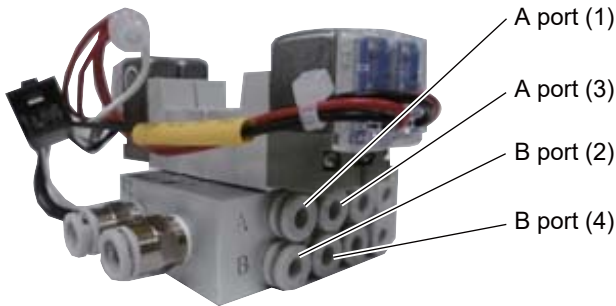


Table 3-1:Hand to solenoid connection

Hand	Hand condition	Solenoid valve port No.	$\Phi 4$ air hose No.	Solenoid valve set in use
Hand 1	Open	A port (1)	1	One set
	Close	B port (2)	2	
Hand 2	Open	A port (3)	3	Two sets
	Close	B port (4)	4	



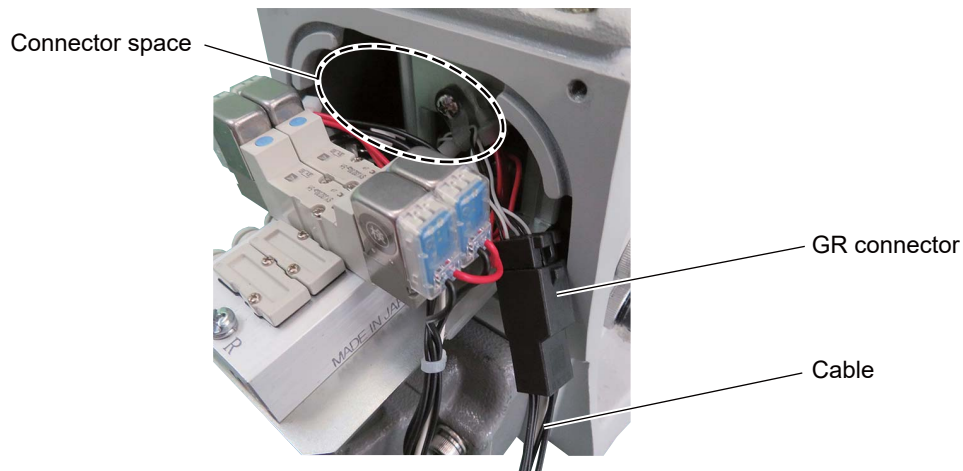
CAUTION

When connecting the $\phi 4$ air hoses, ensure that they are not bent at sharp angles. Ensure that the hoses fit securely inside the robot.

- 6) Connect the hand output cable GR connector to the solenoid valve GR1 connector. After connecting the connectors, locate them in the connector space indicated in the image below.

CAUTION

Ensure that the cables do not come in contact with the surrounding edges.

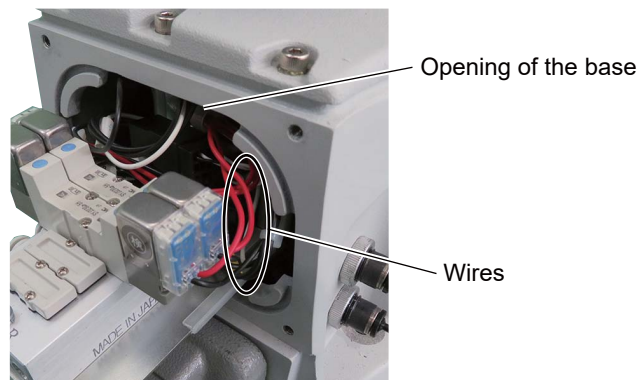


CAUTION

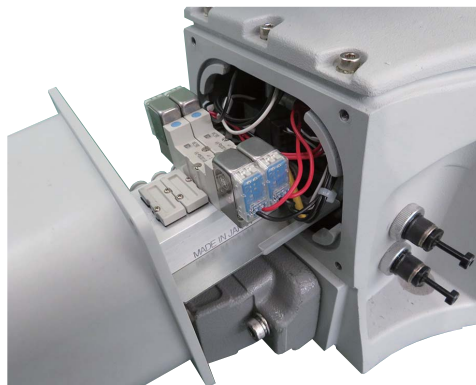
Do not cut the cable tie that holds the cables together in the wiring space.

CAUTION

Ensure that the wires do not protrude beyond the opening of the base.



- 7) Re-install the cover. Then, fix the cover in place using the screws for the cover. (Recommended tightening torque: 1.64 N•m)



CAUTION

Be careful not to damage the gasket when re-installing the cover.

Gasket precautions differ between types A and B because the gasket position differs. The robot will not meet protection specifications if the gasket is not in place or is damaged.

Type A: Insert the gasket into the groove.

Type B: The gasket is attached to the base cover. Check that the gasket is in place.

Acceptable



Type A

Unacceptable



Gasket groove



Type B

- 8) Power on the robot controller. Then select the solenoid valve type (sink/source) and set the parameters "HIOTYPE" and "HANDTYPE" to match the output signals the hand is connected to. For further information on setting parameters, refer to the section "Parameter Settings" in the RT VisualBox Instruction Manual.

4 Maintenance and Inspection

The maintenance and inspection procedures to be carried out to use the robot for a long time without trouble are described in this chapter.

4.1 Type of inspection and maintenance works

Maintenance and inspection are divided into the inspections carried out daily, and the periodic inspections carry out at set intervals. Always carry these out to prevent unforeseen trouble, to maintain the product for a long time, and to secure safety.

[Table 4-1](#) lists the type of inspection and maintenance works.

Table 4-1: Type of inspection and maintenance works

No.	Type of inspection and maintenance works		Description	Operating time ^{Note1)}
1	Daily inspection		Inspection works to be performed every day before starting operation for the safe use of the robot.	-
2	Periodic inspection	Monthly inspection	Inspection and maintenance works to be performed every month.	Every 300hr

Note1) The operating hours assume the robot operation of 15 hours per day for 20 days per month.

When the robot operates for 8 hours per day, the operating hours per month become about a half of the one under the above condition. Then, the monthly inspection is required every two months. To check the periodic inspection schedule and calculate the operating hours, refer to [Page 46, "\(2\) Schedule"](#).

4.2 Inspection items

The inspection items for the robot arm are shown below.

Also refer to section "Maintenance and inspection" in the "Controller Setup and Maintenance" manual, and inspect the controller.

4.2.1 Daily inspection items

Table 4-2 shows the procedure and inspection items. If you notice any abnormal condition, take appropriate measures.

Table 4-2: Daily inspection items (details)

Procedure	Inspection item (details)	Remedies
Before turning power ON (Check the following items before turning the power ON.)		
1	Are any of the robot installation bolts loose? (Visual)	Securely tighten the bolts.
2	Are any of the cover tightening screws loose? (Visual)	Securely tighten the screws.
3	Are any of the hand installation bolts loose? (Visual)	Securely tighten the bolts
4	Is the power supply cable securely connected? (Visual)	Securely connect.
5	Is the machine cable between the robot and controller securely connected? (Visual)	Securely connect.
6	Are there any cracks, foreign contamination or obstacles on the robot and controller cover?	Replace with a new part, or take remedial measures.
7	Is there any abnormality in the pneumatic system? Are there any air leaks, drain clogging or hose damage? Is the air source normal? (Visual)	Drain the drainage, and remedy the air leaks (replace the part).
After turning the power ON (Turn the power ON while monitoring the robot.)		
1	Is there any abnormal motion or abnormal noise when the power is turned ON?	Follow the troubleshooting section.
2	Check that the robot becomes in the servo-off state by the STO function when force is applied to the robot arm while the robot is stopped in the servo-on state. Before using the robot, check that the STR function operates properly.	Contact the manufacturer.
3	Operate the robot at an override speed of 10% or more for roughly 30 seconds. Then check to see if the error "H0216 (Current sensor failure)" has occurred. After that, check that the robot is detecting collision forces properly.	Contact the manufacturer.
During operation (try running with an original program)		
1	Check whether the movement points are deviated? Check the following points if there is any deviation. 1. Are any installation bolts loose? 2. Are any hand installation section bolts loose. 3. Are the positions of the jigs other than the robot deviated? 4. If the positional deviation cannot be corrected, refer to "Troubleshooting", check and remedy.	Follow the troubleshooting section.
2	Is there any abnormal motion or abnormal noise? (Visual)	Follow the troubleshooting section.

4.2.2 Periodic inspection

The inspection items and timings for the robot arm are shown below.

(1) Inspection item

Carry out periodic inspection given in [Table 4-3](#).

Table 4-3: Periodic inspection items (details)

Inspection item (details)	Remedies
Monthly inspection	
Are any of the bolts or screws on the robot arm loose?	Securely tighten the bolts.
Are any of the connector fixing screws or terminal block terminal screws loose?	Securely tighten the screws.

(2) Schedule

The following shows the schedule for the periodic inspection works. Perform the periodic inspection works as appropriate according to the following table.

Operating time Note1)	Inspection schedule		Type of periodic inspection works Note2)
	15 hours per day	8 hours per day	Monthly inspection
300hr	1 month	2 months	○
600hr	2 months	4 months	○
900hr	3 months	6 months	○
1,200hr	4 months	8 months	○
1,500hr	5 months	10 months	○
1,800hr	6 months	12 months	○
:	:	:	:
3,600hr	12 months	24 months	○
:	:	:	:
7,200hr	24 months	48 months	○
:	:	:	:
10,800hr	36 months	72 months	○

Note1) The following shows examples of calculation of the operating hours.

- Operating hours when the robot operates 15 hours per day for 20 days per month for three months:
 $15 \text{ hr/day} \times 20 \text{ days/month} \times 3 \text{ months} = 900 \text{ hr}$
- Operating hours when the robot operates 8 hours per day for 20 days per month for three months:
 $8 \text{ hr/day} \times 20 \text{ days/month} \times 3 \text{ months} = 480 \text{ hr} \dots \text{Approx. } 500 \text{ hr}$

Note2) The item marked with the circle (○) is to be performed. According to the guideline of the operating hours, perform the inspection to check the items described in [Table 4-3](#).

4.3 About Overhaul

Robots which have been in operation for an extended period of time can suffer from wear and other forms of deterioration. In regard to such robots, we define overhaul as an operation to replace parts running out of specified service life or other parts which have been damaged, so that the robots may be put back in shape for continued use. As a rule of thumb, it is recommended that overhaul be carried out before the total amount of servo-on time reaches the specified time (36,000 hours for the robot arm and 36,000 hours for the controller). (See [Fig. 4-1](#).) However, the degree of the equipment's wear and deterioration presumably varies depending on their operating conditions. Especially for operation with high load and frequency, the maintenance cycle may be shorter. For details on the part selection for replacement and the timing of overhaul, contact your dealer.

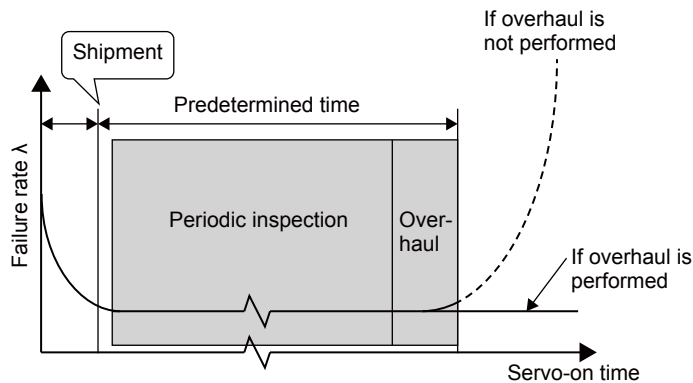


Fig.4-1:Periodic inspection/overhaul periods

4.4 Maintenance parts

Spare parts that are required/recommended for maintenance are indicated in [Table 4-4](#). Only purchase parts directly from specified vendors or the Mitsubishi Electric Service Department.

[Note] Some Mitsubishi-designated parts differ from the maker's standard parts. Thus, confirm the part name, robot arm and controller serial No. and purchase the parts from the dealer.
For replacement work, please contact the dealer.

Table 4-4:Spare parts list

No.	Names	Usage place	Q'ty	Supplier
1	AC servo motor	J1, J2 axis	2	Mitsubishi Electric
2		J3 axis	1	
3		J4, J5 axis	2	
4		J6 axis	1	
5	Reduction gears	J1, J3 axis	2	
6		J2 axis	1	
7		J4 axis	1	
8		J5 axis	1	
9		J6 axis	1	
10	Timing belt	J1 axis	1	
11		J2 axis	1	
12		J3 axis	1	
13		J4 axis	1	
14		J5 axis	1	
15		J6 axis	1	
16	Lithium battery (ER6V)	Base section	1	

4.5 How to release the brake in an emergency

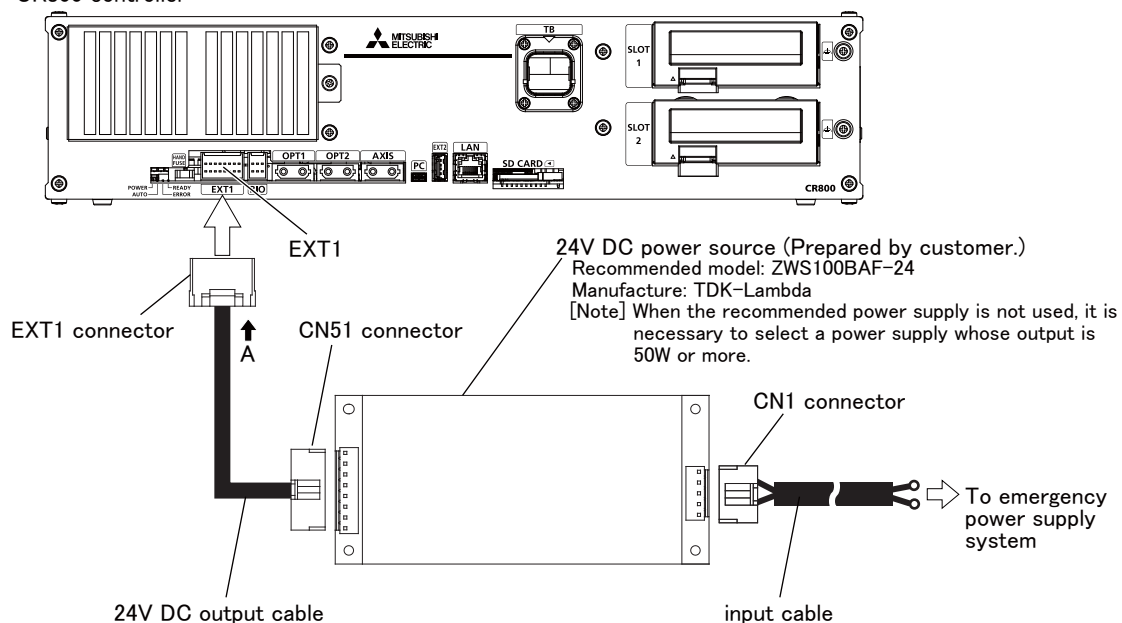
The following describes how to release the brake using the emergency power supply in an emergency. Using an emergency power supply system separated from the primary power supply of the robot prevents the robot from moving accidentally during the brake release operation, which contributes to enhancing customer safety.

An emergency here means an emergency stop of robot operation, which requires the brake release operation to set the robot in an evacuation posture temporarily.

[Note] Do not connect an emergency power supply to multiple robots and release their brakes at the same time.

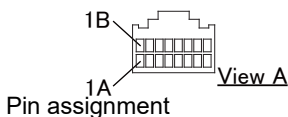
- 1) Connect the teaching pendant to the robot controller.
- 2) Connect the 24V DC power supply to EXT1 of the controller as shown in the figure below.
To prevent accidental power supply from the primary power supply of the robot, prepare an emergency power supply system separated from the primary power supply of the robot to supply AC power to the 24V DC power supply.

CR800 controller



The following shows the connector specifications.

EXT1 connector
Connector: J21DF-16V-KX
Contact: SJ2F-01GF-P1.0
Manufacture: JST



Pin assignment

Pin number	Signal name
8A	24V
8B	
3B	
7A	GND
7B	
3A	

CN51 connector (recommended)
Connector: VHR-8N
Contact: SVH-41T-P1.1
Manufacture: JST

Pin assignment

Pin number	Signal name
1	GND
2	
3	
4	24V
5	
6	
7	
8	

CN1 connector (recommended)
Connector: VHR-5N
Contact: SVH-41T-P1.1
Manufacture: JST

Pin assignment

Pin number	Signal name
1	L
3	N
5	FG

- 3) When the 24V DC power supply is turned on, the controller starts and then errors (H0712, H0090, H0212) will occur.
- 4) Use the teaching pendant or RT VisualBox to release the brakes.
For the brake release operation using the teaching pendant, refer to "Explanation of operation methods" in the separate volume, "Instruction Manual/Detailed Explanations of Functions and Operations".
For further information on releasing the brakes using RT VisualBox, refer to the "Maintenance" section in the RT VisualBox Instruction Manual.

5 Appendix

Appendix 1: How to take out the battery at the disposal of the robot

WARNING

Always perform this procedure with the controller control power turned off.
Failure to observe this may result in electric shock or malfunction.

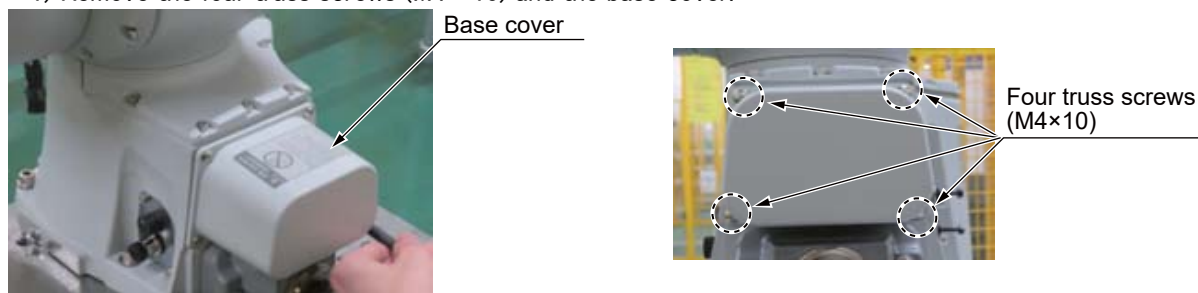
The robot has a battery inside it.

When disposing of the robot, take out the battery for correct disposal.

Take out the battery referring to the following steps with figures.

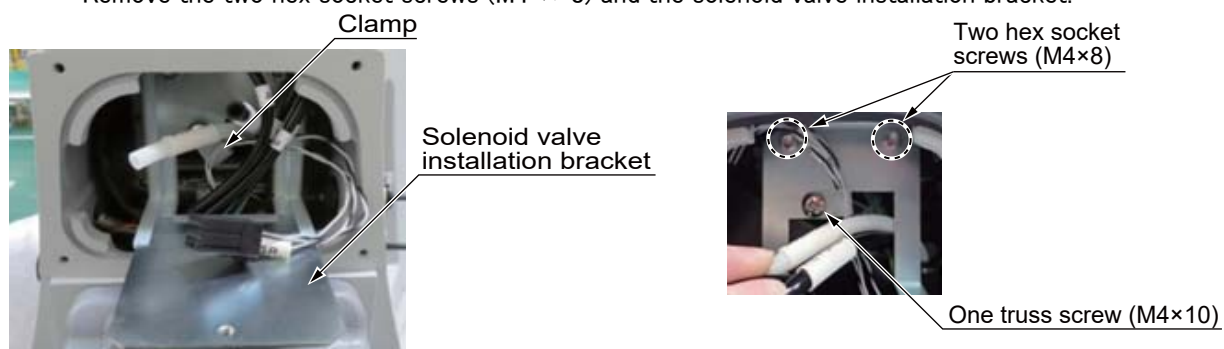
The battery does not need to be replaced during normal use. If a battery replacement is required due to errors or other factors, contact the dealer.

- 1) Remove the four truss screws (M4 × 10) and the base cover.

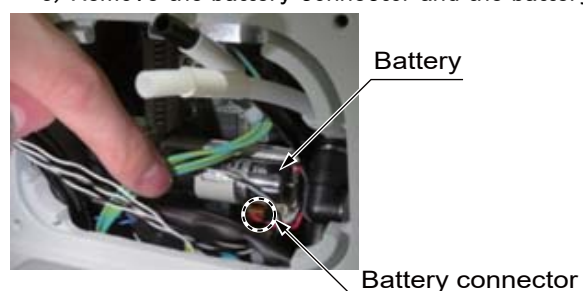


- 2) Remove the one truss screw (M4 × 10) and the clamp securing the cable.

Remove the two hex socket screws (M4 × 8) and the solenoid valve installation bracket.



- 3) Remove the battery connector and the battery.



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